Sec: Sr. IIT **Date: 17-02-24**

Daily Practices Sheet-

MATHS

1. Let S be the set of real numbers. For $a, b \in S$, relation R is defined by $a R b \Leftrightarrow |a-b|<1$ then R is

1) only reflexive
2) only symmetric
3) only transitive
4) Reflexive & symmetric

2. Given that matrix $A=\begin{bmatrix} x & 3 & 2 \\ 1 & y & 4 \\ 2 & 2 & z \end{bmatrix}$. If $xyz=2013$ and $8x+4y+3z=2012$ then $A(adj\, A)$ is equal to

1) $\begin{bmatrix} 68 & 0 & 0 \\ 0 & 68 & 0 \\ 0 & 0 & 68 \end{bmatrix}$
2) $\begin{bmatrix} 34 & 0 & 0 \\ 0 & 34 & 0 \\ 0 & 0 & 34 \end{bmatrix}$
3) $\begin{bmatrix} 29 & 0 & 0 \\ 0 & 29 & 0 \\ 0 & 0 & 29 \end{bmatrix}$
4) $\begin{bmatrix} 58 & 0 & 0 \\ 0 & 58 & 0 \\ 0 & 0 & 58 \end{bmatrix}$

3. In a ΔPQR, if $3\sin P+4\cos Q=6$ and $4\sin Q+3\cos P=1$ then angle 'R' is

1) $\frac{\pi}{6}$
2) $\frac{\pi}{4}$
3) $\frac{3\pi}{4}$
4) $\frac{5\pi}{6}$

4. $\theta \in [0,2\pi]$ and z_1, z_2, z_3 are three complex numbers such that they are collinear and $(1+|\sin\theta|)z_1+(|\cos\theta|-1)z_2-\sqrt{2}\ z_3=0$ .If at least one of the complex numbers z_1, z_2, z_3 is non- zero then number of possible values of θ is

1) Infinite
2) 0
3) 2
4) 4

5. If $\left(1+x+x^2+x^3\right)^{10}=a_0+a_1x+a_2x^2+a_3x^3+\ldots\ldots+a_{30}x^{30}$ then which of the following is true?

1) $a_2-a_4+a_6-\ldots\ldots-a_{28}+a_{30}=1$
2) $a_1-a_3+a_5-a_7+\ldots\ldots\ldots-a_{27}+a_{29}=0$
3) $a_0-a_1+a_2-a_3+\ldots\ldots\ldots\ldots+a_{30}=2^{19}$
4) Both (1) and (2)

6. Let $f(x)=2x^n+\lambda,\ \lambda \in R\, and\ n \in N\ \ f(4)=133$ and $f(5)=255$ then sum of all the positive integral divisors of $(f(3)-f(2))$ is

1) 20
2) 60
3) 21
4) 59

7. In a sequence of 21 terms the first 11 terms are in A.P. with common difference 2 and the last 11 terms are in G.P. with common ratio 2. If the middle term of the A.P. is equal to the middle term of the G.P., then the middle term of the entire sequence is

1) $\frac{-10}{31}$
2) $\frac{-630}{31}$
3) $\frac{-32}{31}$
4) $\frac{-320}{31}$

8. The equation of the plane containing the line $\frac{x-1}{2}=\frac{y+1}{-1}=\frac{z-3}{4}$ and perpendicular to the plane $x+2y+z=12$ is

1) $9x+2y-5z+4=0$
2) $9x-2y+5z-4=0$
3) $9x-2y-5z-4=0$
4) $9x-2y-5z+4=0$

9. The chord of contact of the pair of tangents to the circle $x^2+y^2=1$ drawn from any point on the line $2x+y=4$ always passes through the point

1) $\left(\frac{1}{2},\frac{1}{4}\right)$ 2) $\left(\frac{1}{4},\frac{1}{2}\right)$ 3) $\left(1,\frac{1}{2}\right)$ 4) $\left(\frac{1}{2},1\right)$

10. The number of real common normals of $y^2 = 4x$ and $x^2 = 4y$ that can be drawn is equal to

1) 1 2) 3 3) 5 4) 6

11. A variable triangle ABC is inscribed in a circle of diameter x units. At a particular instant the rate of change of side 'a' is $x/2$ times the rate of change of the opposite angle A then $A =$

1) $\pi/6$ 2) $\pi/3$ 3) $\pi/4$ 4) $\pi/2$

12. The line $3x+2y=24$ meets x-axis at A and y-axis at B. The perpendicular bisector of $\overline{AB}$ meets the line passing through $(0,-1)$ and parallel to x-axis at 'C'. Then the area of ΔABC is (in square units)

1) 83 2) 87 3) 90 4) 91

13. The area bounded by $y = xe^{|x|}$ and the lines $|x| = 1, y = 0$ is (in sq.units)

1) 1 2) 2 3) 4 4) 6

14. A point (a,b) is called a good point if both 'a' and 'b' are integers. The number of good points on the curve xy= 225 are

1) 9 2) 10 3) 16 4) 18

15. The area bounded by the circles $x^2 + y^2 = 1$ & $x^2 + y^2 = 4$ and the pair of lines $2x^2 - 3xy - 2y^2 = 0$ $(y>0)$ is ___(in sq.units)

1) $\pi/4$ 2) $\pi/2$ 3) $3\pi/4$ 4) $\pi/3$

16. A random variable X has its range $\{0,1,2\}$ and $P(X=0)=3c^3$, $P(X=1)=4c-10c^2$, $P(X=2)=5c-1$ then $P(0<X<3)=$ ____

1) $\frac{1}{3}$ 2) $\frac{8}{9}$ 3) $\frac{7}{8}$ 4) $\frac{1}{9}$

17. A tower ABCD stands on a level ground with foot A. At a point P on the ground, the portion AB, AC and AD subtends angles α,β and γ respectively. If $AB=a, AC=b, AD=c$, $AP=x$ and $\alpha+\beta+\gamma=180^0$ then $(a+b+c)x^2=$

1) abc 2) $\frac{1}{abc}$ 3) $a+b-c$ 4) $a-b-c$

18. Let A be a non – singular matrix such that $A^{-1}=\begin{pmatrix}1 & 1 & 1\\ 4 & 6 & 8\\ 100 & 100 & 99\end{pmatrix}$. Then the value of $\det(adj(2A))+\det(2A)=$

1) 8 2) 12 3) 16 4) 28

19. The equation of the curve passing through $(2,2)$ and satisfying the differential equation $\frac{dy}{dx}+2xy=x^2\left(3-\frac{dy}{dx}\right)$ is

1) $x^2(x-y)=x-2$ 2) $y^2(x-y)=y-2$ 3) $y^2(x-y)=x-2$ 4) $x^2(x-y)=y-2$

20. If α,β,γ are the roots of $x^3-3x^2+3x+7=0$ and ω is a complex cube root of unity, then the value of $\left(\frac{\alpha-1}{\beta-1}+\frac{\beta-1}{\gamma-1}+\frac{\gamma-1}{\alpha-1}\right)$ is equal to $p(\omega^q)$ for some $p,q\in Z$. Then the number of ordered pairs (p,q) such that $p+q=15$ is

1) 0 2) 1 3) 2 4) infinite

21. Let f be a real function satisfying $f(x)=f(x+3)-f(x+6)+f(x+9)$. Then the function

$$g(x)=\int_{x}^{x+12} f(t)\,dt \text{ is}$$

1) a constant function 2) a linear function of x
3) an exponential function of x 4) a pure quadratic function of x

22. **Statement (S_1):** Let $f(x)=\begin{cases} 5x-7 & for\ x\le 2 \\ \int_0^x (1+|1-t|)\,dt & for\ x>2 \end{cases}$

Then f is not differentiable at $x=2$

Statement (S_2): A function $f(x)$ is not differentiable at $x=a$, if f is not continuous at $x=a$, where $f(x)$ is defined in the neighbourhood of $'a'$

1) S_1 is true, S_2 is true and S_2 is correct explanation for S_1
2) S_1 is true, S_2 is false
3) S_1 is true, S_2 is true but S_2 is not a correct explanation for S_1
4) S_1 is false, S_2 is true

23. The sides of the rectangle of the greatest area that can be inscribed in the ellipse $x^2+2y^2=8$ are given by

1) $4\sqrt{2}, 4$ 2) $4, 2\sqrt{2}$ 3) $2, \sqrt{2}$ 4) $2\sqrt{2}, 2$

24. If the points $A(z), B(-z), C(1-z)$ in argand plane are vertices of an equilateral triangle, then $z=$

1) $1\pm i\dfrac{\sqrt{3}}{2}$ 2) $\dfrac{1}{2}(1\pm i)$ 3) $\dfrac{1}{4}(1\pm\sqrt{3}i)$ 4) $\dfrac{1}{3}(1\pm\sqrt{3}i)$

25. **Statement (S_1):** If the straight line given by $\bar{r}=(2\bar{i}-\bar{j})+\lambda(x\bar{i}+2\bar{j}-3\bar{k})$, $\lambda\in R$ and the points with position vectors $\bar{i}+2\bar{j}+3\bar{k}$, $\bar{i}+\bar{j}+\bar{k}$ are coplanar, then the value of x is $-\dfrac{7}{3}$

Statement (S_2): Three vectors $\bar{a}, \bar{b}, \bar{c}$ are coplanar iff $\left[\bar{a}, \bar{b}, \bar{c}\right]=0$

1) S_1 is true, S_2 is true and S_2 is a correct explanation for S_1
2) S_1 is true, S_2 is true and S_2 is not a correct explanation for S_1
3) S_1 is true, S_2 is false 4) S_1 is false, S_2 is true

26. If $(0,1),(1,1)\ and\ (1,0)$ are the mid-points of the sides of ΔABC, then the abscissa of in-centre of ΔABC, is equal to

1) $2\tan\dfrac{\pi}{8}$ 2) $2\tan\dfrac{3\pi}{8}$ 3) $\sqrt{2}\tan\dfrac{\pi}{8}$ 4) $\sqrt{2}\tan\dfrac{3\pi}{8}$

27. If r_1, r_2, r_3 are the ex-radii of ΔABC, then the value of $\begin{vmatrix} r_1 & r_2 & r_3 \\ r_2 & r_3 & r_1 \\ r_3 & r_1 & r_2 \end{vmatrix}$ is

1) Positive 2) Negative
3) Non-positive 4) Non-negative

28. The number of 4 – digit natural numbers, such that the sum of the digits of each number is equal to 9 is

1) 165 2) 210 3) 265 4) None of these

29. $$\int \frac{x+x^{2/3}+x^{1/6}}{x\left(1+x^{1/3}\right)}dx=$$

1) $\frac{3}{2}x^{2/3}+\tan^{-1}\left(x^{1/6}\right)+c$ 2) $\frac{2}{3}x^{2/3}+6\tan^{-1}\left(x^{1/6}\right)+c$

3) $\frac{3}{2}x^{2/3}-6\tan^{-1}\left(x^{1/6}\right)+c$ 4) $\frac{3}{2}x^{2/3}+6\tan^{-1}\left(x^{1/6}\right)+c$

30. If $\int_0^x f(t)dt = x+\int_x^1 tf(t)\,dt$, then the value of $f(1)$ is

1) $\frac{1}{2}$ 2) 0 3) 1 4) $-\frac{1}{2}$

PHYSICS

31. When a body is suspended from a fixed point by a spring, the angular frequency of its vertical oscillations is ω_1. When a different spring is used, the angular frequency is ω_2. The angular frequency of vertical oscillations when both the springs are used together in series is given by

1) $\omega=\left[\omega_1^2+\omega_2^2\right]^{\frac{1}{2}}$ 2) $\omega=\left[\frac{\omega_1^2+\omega_2^2}{2}\right]^{\frac{1}{2}}$ 3) $\omega=\left[\frac{\omega_1^2\omega_2^2}{\omega_1^2+\omega_2^2}\right]^{\frac{1}{2}}$ 4) $\omega=\left[\frac{\omega_1^2\omega_2^2}{2\left(\omega_1^2+\omega_2^2\right)}\right]^{\frac{1}{2}}$

32. A small pond of depth 0.5m deep is exposed to a cold winter with outside temperature of 263K. Thermal conductivity of ice is $K=2.2W\,m^{-1}K^{-1}$, latent heat $L=3.4\times10^5\ Jkg^{-1}$ and density $\rho=0.9\times10^3$. Take the temperature of the pond to be 273 K. The time taken for the whole pond to freeze is about.

1) 20 days 2) 25 days 3) 30 days 4) 35 days

33. A body of mass 4 kg moves under the action of a force $\vec{F}=\left(4\hat{i}+12t^2\hat{j}\right)N$. where t is the time in second. The initial velocity of the particle is $\left(2\hat{i}+\hat{j}+2\hat{k}\right)ms^{-1}$. If the force is applied for 1 s, work done is :

1) 4 J 2) 8 J 3) 12 J 4) 16 J

34. A racing car moves along circular track of radius b. The car starts from rest and its speed increases at a constant rate α. Let the angle between the velocity and the acceleration be θ at time t. Then $(\cos\theta)$ is

1) 0 2) $\alpha t^2/b$ 3) $\frac{b}{\left(b+\alpha t^2\right)}$ 4) $\frac{b}{\left(b^2+\alpha^2t^4\right)^{\frac{1}{2}}}$

35. A small fish, 4cm below the surface of a lake, is viewed through a thin converging lens of focal length 30 cm held 2 cm above the water surface. Refractive index of water is 1.33. The image of the fish from the lens is at a distance of

1) 10 cm 2) 8 cm 3) 6 cm 4) 4 cm

36. An isolated metallic object is charged in vacuum to a potential V_0 using a suitable source, its electrostatic energy being W_0. It is then disconnected from the source and immersed in a large volume of dielectric with dielectric constant K. The electrostatic energy of the sphere in the dielectric is :

1) K^2W_0 2) KW_0 3) $\frac{W_0}{K^2}$ 4) $\frac{W_0}{K}$

37. In cases of real images formed by a thin convex lens, the linear magnification is (I) directly proportional to the image distance, (II) inversely proportional to the object distance, (III) directly proportional to the distance of image from the nearest principal focus, (IV) inversely proportional to the distance of the object from the nearest principal focus. From these the correct statements are :

1) (I) and (II) only 2) (III) and (IV) only

3) (I) , (II), (III) and (IV) all 4) None of (I), (II), (III) and (IV)

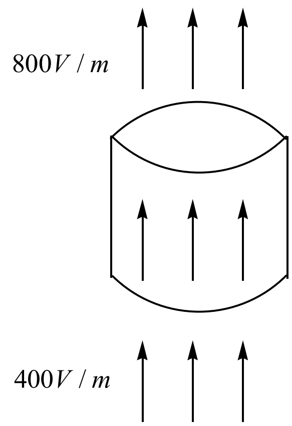
38. A quantity α is defined as $\alpha=\frac{e^2}{4\pi\varepsilon_0 ch}$. where e is electric charge, $h=\frac{h}{2\pi}$ is the reduced Planck's constant and c is the speed of light. The dimensions of α are

1) $\left[M^0L^0T^0I^0\right]$ 2) $\left[M^1L^{-1}T^2I^{-2}\right]$ 3) $\left[M^2L^1T^{-1}I^0\right]$ 4) $\left[M^0L^3T^{-1}I^{-2}\right]$

39. A slit of width a is illuminated by parallel monochromatic light of wavelength λ. The value of a at which the first minimum of the diffraction pattern will form at $\theta = 30^0$ is

1) $\lambda/2$ 2) λ 3) 2λ 4) 3λ

40. A cylinder on whose surfaces there is a vertical electric field of varying magnitude as shown. The electric field is uniform on the top surface as well as on the bottom surface therefore, this cylinder encloses

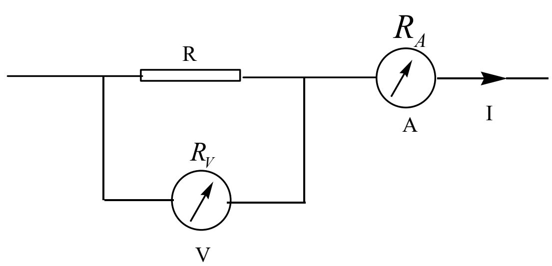


1) no net charge
2) net positive charge
3) net negative charge
4) There is not enough information to determine whether or not there is net charge inside the cylinder.

41. A coaxial cable consists of two thin cylindrical conducting shells of radii a and b $(a < b)$. The inductance per unit length of the cable is

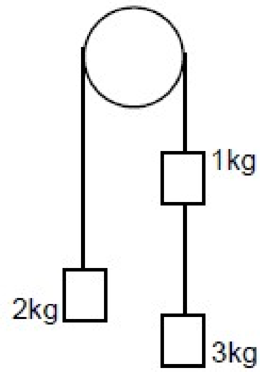
1) $\frac{\mu_0}{2\pi}\frac{(a-b)}{a}$ 2) $\frac{\mu_0}{4\pi}\ln\left(\frac{a}{b}\right)$ 3) $\frac{\mu_0}{4\pi}\ln\left(\frac{b}{a}\right)$ 4) $\frac{\mu_0}{2\pi}\ln\left(\frac{b}{a}\right)$

42. Let V and I be the readings of the voltmeter and the ammeter respectively as shown in the figure. Let R_V and R_A be their corresponding resistance, therefore



1) $R = \frac{V}{I}$ 2) $R = \frac{V}{I - \left(\frac{V}{R_V}\right)}$ 3) $R = R_V - R_A$ 4) $R = \frac{V(R+R_A)}{IR_A}$

43. In the following arrangement the pulley is assumed to be light and the string inextensible. The acceleration of the system can be determined by considering conservation of a certain physical quantity. The physical quantity conserved and the acceleration respectively, are



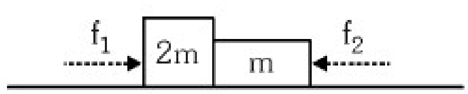
1) energy and 2g/3 2) linear momentum and g/3
3) angular momentum and g/3 4) none

44. Two moles of hydrogen are mixed with n moles of helium. The root mean square speed of gas molecules in the mixture is $\sqrt{2}$ times the speed of sound in the mixture. Then n is.
1) 3 2) 2 3) 1.5 4) 2.5

45. A wire in the shape of square frame carries a current I and produces a magnetic field B_S at its centre. Now the wire is bent in the shape of a circle and carries the same current. If B_C is the magnetic field produced at the centre of the circular coil, then B_S / B_C is
1) $8\pi^2$ 2) $\frac{8\pi^2}{\sqrt{2}}$ 3) $\frac{8\sqrt{2}}{\pi^2}$ 4) $8\pi\sqrt{2}$

46. Ultraviolet light of wavelength 300 nm and intensity 1 W/ m^2 falls on the surface of a photosensitive material. If one percent of the incident photons produce photoelectrons then the number of photoelectrons emitted per second from and area of $1cm^2$ of the surface is nearly
1) 1.51×10^{13} 2) 1.51×10^{12} 3) 4.12×10^{13} 4) 2.13×10^{11}

47. Two blocks of masses m and 2m are placed on a smooth horizontal surface as shown. In the first case only f_1 is applied from left. Later on only a force f_2 is applied from right. If the force acting at the interface of the two blocks in the two cases is the same, then $f_1 : f_2$ is



1) 1:1 2) 1:2 3) 2:1 4) 1:3

48. A ball A of mass 1 kg moving at a speed of 5 m/s strikes tangentially another ball B initially at rest. The ball A then moves at right angles to its initial direction at a speed of 4 m/s . If the collision is elastic , the mass (in kg) of ball B and its momentum after collision (in kg – m/s) respectively are approximately)
1) 1.2 and 1.8 2) 2.2 and 3.3 3) 4.6 and 6.4 4) 6.2 and 9.1

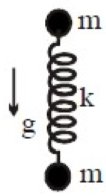
49. Two identical rooms in a house are connected by an open doorway. The temperatures in the two rooms are maintained at two different values. Therefore,
1) The room with higher temperature contains more amount of air.
2) The room with lower temperature contains more amount of air.
3) Both the rooms contain the same amount of air.
4) The room with higher pressure contains more amount of air.

50. A vibrator of frequency f is placed near one end of a long cylindrical tube. The tube is fitted with a movable piston at the other end. An observer listens to the sound through a side opening. As the piston is moved through 8.75 cm, the intensity of sound recorded by the observer changes from a maximum to a minimum. If the speed of sound in air is 350 m/s, the frequency f is
1) 500Hz 2) 1000 Hz 3) 2000 Hz 4) 4000 Hz

51. A heavy metal block is dragged along a rough horizontal surface at a constant speed of 20 km/hr. The coefficient of friction between the block and the surface is 0.6. The block is made of a material whose specific heat is $0.1\ \text{cal/g}\ ^\circ C$ and absorbs 25% of heat generated due to friction. If the block is dragged for 10 min, the rise in temperature of the block is about $(g = 10\ m/s^2)$

1) $12^0 C$ 2) $50\ ^0C$ 3) $210\ ^0C$ 4) Data insufficient

52. A mass m is hung on an ideal massless spring. Another equal mass is connected to the other end of the spring. The whole system is at rest. At t = 0, m is released and the system falls freely under gravity. Assume that natural length of the spring is L_0 its initial stretched length is L and the acceleration due to gravity is g. What is distance between masses as function of time ?



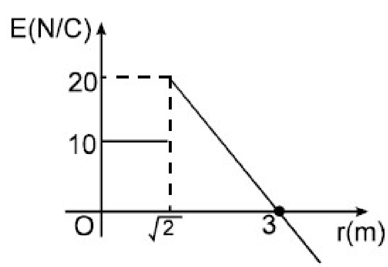
1) $L_0 + (L - L_0)\cos\sqrt{\frac{2k}{m}}t$ 2) $L_0 + (L - L_0)\cos\sqrt{\frac{k}{m}}t$

3) $L_0 - 2(L + 2L_0)\cos\sqrt{\frac{2k}{m}}t$ 4) $L_0 + (L - L_0)\sin\sqrt{\frac{2k}{m}}t$

53. An upolarized light of intensity 32 W/ m^2 passes through three polarizers, such that the transmission axis of last polarizer is crossed with that of the first. If the intensity of emergent light is $3W/m^2$, then the angle between the transmission axes of the first two polarizers is

1) 30^0 2) 19^0 3) 45^0 4) 90^0

54. An electric field 'E' whose direction is radially outward varies as distance from origin 'r' as shown in the graph. E is taken as positive if its direction is away from the origin. Then the work done by electric field on a 2 C charge if it is taken from $(1,\ 1,\ 0)$ to $(3,0,0)$ is :



1) $20(3-\sqrt{2})J$ 2) $-60J$ 3) $60J$ 4) $20(\sqrt{2}-3)J$

55. A particle of mass 10 g starts from rest at t=0 s from a point (0 m, 4 m) and gets accelerated at $0.5 m/s^2$ along $x - \sqrt{3}y + 4\sqrt{3} = 0$ in XY plane. The angular momentum of the particle about the origin (in SI units) at t = 2 s is

1) $-0.01\sqrt{3}\hat{k}$ 2) $-0.02\sqrt{3}\hat{k}$ 3) zero 4) $-20\sqrt{3}\hat{k}$

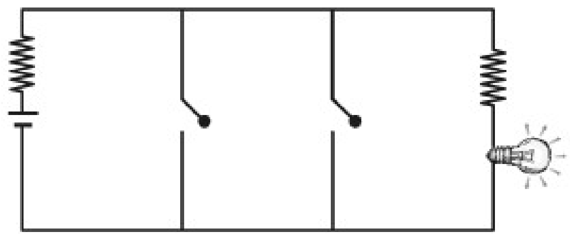
56. A thin rod of length l in the shape of a semicircle is pivoted at one of its ends such that it is free to oscillate in its own place. The frequency f of small oscillations of the semicircular rod is

1) $\frac{1}{2\pi}\sqrt{\frac{g\pi}{2l}}$ 2) $\frac{1}{2\pi}\sqrt{\frac{g\sqrt{\pi^2+4}}{2l}}$ 3) $\frac{1}{2\pi}\sqrt{\frac{\sqrt{g(\pi+2)}}{l}}$ 4) $\frac{1}{2\pi}\sqrt{\frac{g(\pi^2+1)}{2\pi l}}$

57. When a light wave is incident at the interface between two media, the reflection coefficient is given by $\frac{(n-1)^2}{(n+1)^2}$ where n is the refractive index of the denser medium with respect to the rarer medium. Two stretched strings whose linear densities are 25 g/m and 9 g/m are joined together. Assuming the law of optics holds good here also , the reflection coefficient for the pulse along the strings is

1) $\frac{9}{16}$ 2) $\frac{3}{4}$ 3) $\frac{1}{16}$ 4) $\frac{1}{9}$

58. The circuit shown below is equivalent to

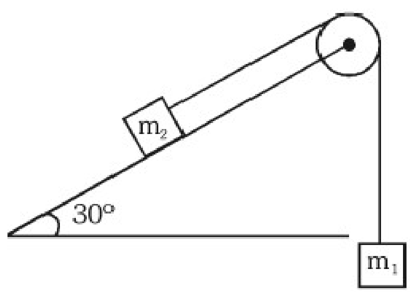


1) OR gate 2) NOR gate 3) AND gate 4) NAND gate

59. A charge (-2Q) is distributed uniformly on a spherical balloon of radius R. Another point charge (+Q) is situated at the centre of the balloon. The balloon is now inflated to twice the radius. Neglecting the elastic energy involved in the process , the change in total electric energy of the system is

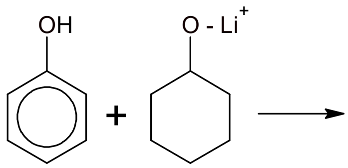
1) $\frac{-Q^2}{2\pi \in_0 R}$ 2) $\frac{-Q^2}{4\pi \in_0 R}$ 3) $\frac{+Q^2}{4\pi \in_0 R}$ 4)Zero

60. Masses m_1 *and* m_2 are connected to a string passing a pulley as shown, Mass m_1 starts from rest and falls through a distance d in time t. Now, by inter charging the masses the time required for m_2 to fall through the same distance is 2t . Therefore, the ratio of masses $m_1 : m_2$ is

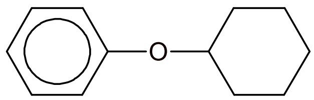


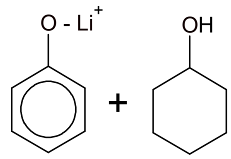
1) $\frac{2}{3}$ 2) $\frac{3}{2}$ 3) $\frac{5}{2}$ 4) $\frac{4}{3}$

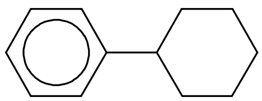
CHEMISTRY

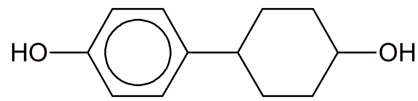
61. 

What is/are the product(s) of the above reaction?

1) 

2) 

3) 

4) 

62. Identify the correct statement of the following

A) Hypo forms super saturated solutions

B) on thermal decomposition hypo gives H_2S, SO_2 and S.

C) Dilute sodium thio sulphate, on reaction with $AgNO_3$ finally gives black ppt. of Ag_2S

D) AgBr can be used in making photo graphic films.

1) A, B, C 2) B, C, D 3) A, C, D 4) A, B, C, D

63. Specify the coordination geometry around and hybridisation of N and B atoms in a 1 : 1 complex of BF_3 and NH_3

1) N : tetrahedral, sp^3 ; B–tetrahedral, sp^3

2) N–Pyramidal, sp^3; B–pyramidal, sp^3

3) N–pyramidal, sp^3; B–planar, sp^2

4) N–pyramidal, sp^3; B–tetrahedral – sp^3

64. Two components A and B form an ideal solution. The mole fractions of A and B in ideal solution are X_A and X_B, while that of in vapour phase, these components have their mole fractions as Y_A and Y_B. Then, the slope and intercept of plot of $\frac{1}{Y_A} vs \frac{1}{X_A}$ will be :

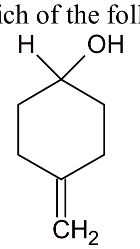
1) $\frac{P_A^0}{P_B^0}, \frac{P_B^0 - P_A^0}{P_B^0}$

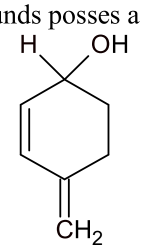
2) $\frac{P_B^0}{P_A^0} . \frac{P_A^0 - P_B^0}{P_A^0}$

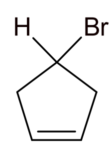
.3) $\frac{P_B^0}{P_A^0}, \frac{P_B^0}{P_3^0 - P_A^0}$

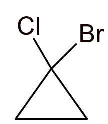
4) $P_A^0 - P_B^0, \frac{P_A^0}{P_B^0}$

65. Which of the following compounds posses a chiral centre?

1) 

2) 

3) 

4) 

66. Which of the following has the minimum heat of dissociation:

1) $[(CH_3)_3 N \rightarrow BF_3]$

2) $[(CH_3)_3 N \rightarrow B(CH_3)F_2]$

3) $[(CH_3)_3 N \rightarrow B(CH_3)_2 F]$

4) $[(CH_3)_3 N \rightarrow B(CH_3)_3]$

67. Which among the following is most soluble in water?

1) $CsClO_4$ 2) $NaClO_4$ 3) $LiClO_4$ 4) $KClO_4$

68. For Adiabatic free expansion $(P_{ext} = 0)$ of an ideal gas

1) $\Delta S_{surrounding} = 0$

2) $\Delta S_{surrounding} < 0$

3) $\Delta S_{surrounding} > 0$

4) $\Delta S_{system} = 0$

69. $C_6H_5 - CO - CH_2 - CH_2 - CH_2 - COOH \rightarrow C_6H_5 - CO - CH_2 - CH_2 - CH_2 - CD_2OH$.

This conversion is done by

1) $NaBH_4 / H_3O^+$ followed by $LiAlD_4 /\ H_2O$
2) $LiAlD_4 /\ H_2O$ followed by $NaBH_4 / H_3O^+$
3) $(CH_2OH)_2$ followed by $LiAlD_4 / H_3O^+$
4) $DMgBr\ / H_3O^+$

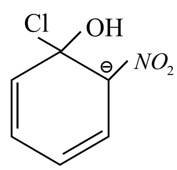
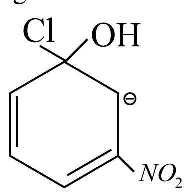
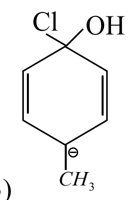
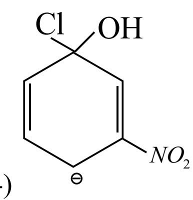
70. On heating potassium ferrocyanide with conc. H_2SO_4 produces a neutral gas ‘A’. The gas ‘A’ on treatment with caustic soda under high pressure produced ‘B’, what are ‘A’ and ‘B’ respectively.

1) CO_2, Na_2CO_3 2) SO_2, Na_2SO_4
3) CO, HCOONa 4) NO_2, $NaNO_3$

71. Among the following statements, the incorrect one is :

1) Calamine and siderite are carbonates 2) Argentite and cuprite are oxides
3) Zinc blende and pyrites are sulphides 4) Malachite and azurite are ores of copper

72. Which of the following transition state is more stable in the nucleophilic substitution ?

1)  2)  3)  4) 

73. An organic compound upon hydrolysis produces two compounds one product gave silver mirror test, other product reacts with Hinsberg reagent to produce an alkali insoluble product. The organic compound is

1) $CH_3-CH_2-\overset{\overset{O}{||}}{C}-NHCH_3$ 2) $H-\overset{\overset{O}{||}}{C}-N\begin{matrix}CH_3\\CH_3\end{matrix}$

3) $CH_3-CH_2-\overset{\overset{O}{||}}{C}-NH_3$ 4) $H-\overset{\overset{O}{||}}{C}-NH-CH_2-CH_3$

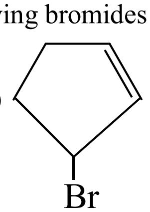
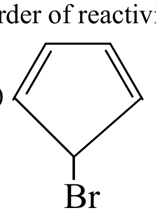
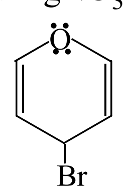
74. 100 ml of a sample of hard water after passing through cation exchange resin, required 20ml of 0.05M NaOH for neutralisation. One litre of same sample of water on treatment with sufficient lime gave 200mg of $CaCO_3$. Assuming that the hardness is only due to Ca^{+2} ions. Find the degree of permanent hardness of water.

1)300ppm 2)150ppm 3) 100ppm 4) 200ppm

75. Calculate the pH at which $Mg(OH)_2$ begins to precipitate from solution containing 0.1 M Mg^{+2} ions. Ksp for $Mg(OH)_2$ is 1.0×10^{-11}

1) 4 2) 9 3) 5 4) 8

76. Arrange the following bromides in the increasing order of reactivity towards $AgNO_3$

A)  B)  C) 

1) $C < A < B$ 2) $C > A > B$ 3) $B < A < C$ 4) $A < B < C$

77. Shape of MnO_4^- and hybridization of Mn in MnO_4^- are respectively

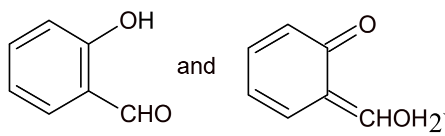
1) Tetrahedral, sp^3 2) Tetrahedral, d^3s

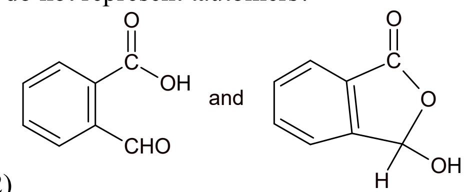
3) Sq. planar, dsp^2 4) sq. planar sp^2d

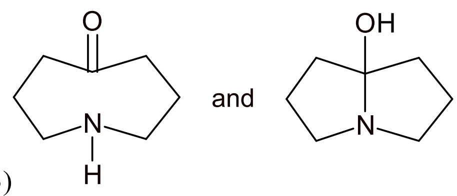
78. Bond dissociation energy of XY, X_2 and Y_2 (all diatomic molecules) are in the ratio 1:1: 0.5 and ΔH_f of XY is $-200\ kJ\ mol^{-1}$. The bond dissociation energy of X_2 will be:

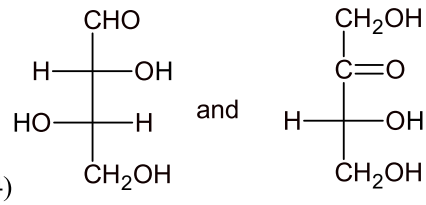
1) $800\ kJ\ mol^{-1}$ 2) $200\ kJ\ mol^{-1}$ 3) $300\ kJ\ mol^{-1}$ 4) $400\ kJ\ mol^{-1}$

79. Which of the following pairs of structures do not represent tautomers?

1) 

2) 

3) 

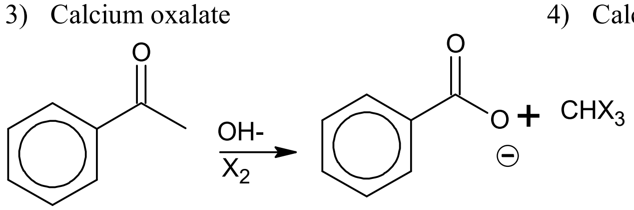
4) 

80. Consider a titration of potassium dichromate solution with acidified Mohr's salt solution using diphenylamine as indicator. The number of moles of Mohr's salt required per mole of dichromate is

1) 3 2) 4 3)5 4)6

81. The compound insoluble in acetic acid is

1) Calcium oxide 2) Calcium carbonate
3) Calcium oxalate 4) Calcium hydroxide



82. Which of the following is correct comparison of rate of haloform reaction with various halogens ?

1) $rCl_2 > rBr_2 > rI_2$ 2) $rI_2 > rBr_2 > rCl_2$
3) $rBr_2 > rCl_2 > rI_2$ 4) $rCl_2 \approx rBr_2 \approx rI_2$

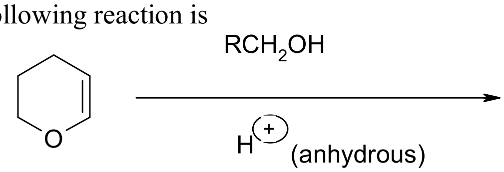
83. A complex of certain metal has the magnetic moment of 4.91 BM whereas another complex of the same metal with same oxidation state has zero magnetic moment. The metal ion could be…… and if that metal ion forms a complex with EDTA, then its EAN would be……

1) Co^{2+},36 2) Mn^{2+},38 3) Fe^{2+},36 4) Ag^{+},36

84. Of the following metals, which one cannot be obtained by electrolysis of the aqueous solution of its salt is

1) Ag 2)Mg 3) Cu 4)Au

85. The major product of the following reaction is



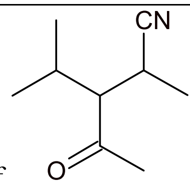
1) A hemiacetal 2) An acetal 3) An ether 4) An ester

86. The compressibility of a gas is less than unity at STP. Therefore,

1) $V_m > 22.4\,L$ 2) $V_m < 22.4\,L$ 3) $V_m = 22.4\,L$ 4) $V_m = 44.8\,L$

87. Nitrogen dioxide cannot be obtained by heating

1) KNO_3 2) $Pb(NO_3)_2$ 3) $Cu(NO_3)_2$ 4) $AgNO_3$

88. The IUPAC name of  is
1) 2-methyl-3-(1-methylethyl)-4-oxopentanenitrile
2) 4-cyano–3-(1-methylethyl)–2–pentanone
3) 3-acetyl–2–cyano–4–methylpentane
4) 3–ethanoyl–2–methyl–3–(1–methylethyl) pentanenitrile

89. For the electrochemical cell, $M\left|M^{+}\right\|X^{-}\left|X\right.$, $E^{\circ}\left(M^{+}|M\right)=0.44\text{ V}$ and $E^{\circ}\left(X/X^{-}\right)=0.33\text{ V}$. From this data one can deduce that
1) $M+X\rightarrow M^{+}+X^{-}$ is the spontaneous reaction
2) $M^{+}+X^{-}\rightarrow M+X$ is the spontaneous reaction
3) $E_{cell}=0.77\text{ V}$
4) $E_{cell}=-0.77\text{ V}$

90. The exhausted permutit is generally regenerated by percolating through it a solution of:
1) Sodium chloride 2) Calcium chloride
3) Magnesium chloride 4) Potassium chloride

1	**4**	2	**3**	3	**1**	4	**4**	5	**4**
6	**2**	7	**1**	8	**4**	9	**1**	10	**1**
11	**2**	12	**4**	13	**2**	14	**4**	15	**3**
16	**2**	17	**1**	18	**2**	19	**4**	20	**1**
21	**1**	22	**3**	23	**2**	24	**3**	25	**1**
26	**3**	27	**3**	28	**1**	29	**4**	30	**1**
31	**3**	32	**1**	33	**4**	34	**4**	35	**3**
36	**4**	37	**2**	38	**1**	39	**3**	40	**2**
41	**4**	42	**2**	43	**1**	44	**2**	45	**3**
46	**2**	47	**3**	48	**3**	49	**2**	50	**2**
51	**1**	52	**1**	53	**1**	54	**1**	55	**2**
56	**2**	57	**3**	58	**2**	59	**4**	60	**2**
61	**2**	62	**4**	63	**1**	64	**2**	65	**2**
66	**4**	67	**3**	68	**1**	69	**3**	70	**3**
71	**2**	72	**1**	73	**2**	74	**1**	75	**2**
76	**3**	77	**2**	78	**4**	79	**4**	80	**4**
81	**3**	82	**4**	83	**3**	84	**2**	85	**2**
86	**2**	87	**1**	88	**1**	89	**2**	90	**1**

Sec: Sr. IIT **Date: 19-02-24**

Daily Practices Sheet - 02

MATHS

1. $\int \frac{(2+\sqrt{x})dx}{(x+1+\sqrt{x})^2}$ is equal to:

1) $\frac{x}{x+\sqrt{x}+1}+C$ 2) $\frac{2x}{x+\sqrt{x}+1}+C$ 3) $\frac{-2x}{x+\sqrt{x}+1}+C$ 4) $\frac{-x}{x+\sqrt{x}+1}+C$

2. $f(x)=\frac{x}{1+(\ln x)(\ln x).....\infty}, \forall x \in [1,\infty)$ then $\int_1^{2e} f(x)dx$ equal is :

1) $\frac{e^2-1}{2}$ 2) $\frac{e^2+1}{2}$ 3) $\frac{e^2-2e}{2}$ 4) e^2-1

3. Let $L=\lim_{n\to\infty}\left(\frac{(2.1+n)}{1^2+n.1+n^2}+\frac{(2.2+n)}{2^2+n.2+n^2}+\frac{(2.3+n)}{3^2+n.3+n^2}+....+\frac{(2.n+n)}{3n^2}\right)$ then value of e^L is :

1) 2 2) 3 3) 4 4) 3/2

4. The minimum value of $f(x)=\int_0^4 e^{|x-t|}dt$,*if* $x\in[0,3]$ is :

1) $2e^2-1$ 2) e^4-1 3) $2(e^2-1)$ 4) e^2-1

5. Let z be a complex number satisfying $|z-3|\le|z-1|, |z-3|\le|z-5|, |z-i|\le|z+i|$ and $|z-i|\le|z-5i|$. Then the area of region in which z lies is A square units, where A = :

1) 3 2) 2 3) 4 4) 6

6. Let $f(x)$ be differentiable function on the interval $(0,\infty)$ such that $f(1)=1$ and $\lim_{t\to x}\left(\frac{t^3 f(x)-x^3 f(t)}{t^2-x^2}\right)=\frac{1}{2}, \forall x>0,$ then $f(x)$ is :

1) $\frac{1}{4x}+\frac{3x^2}{4}$ 2) $\frac{3}{4x}+\frac{x^3}{4}$ 3) $\frac{1}{4x}+\frac{3x^3}{4}$ 4) $\frac{1}{4x^3}+\frac{3x}{4}$

7. If $a,b\in R$ distinct numbers satisfying $|a-1|+|b-1|=|a|+|b|=|a+1|+|b+1|$, then the minimum value of $|a-b|$ is :

1) 3 2) 0 3) 1 4) 2

8. If α,β are the roots of the quadratic equation $x^2-\left(3+2^{\sqrt{\log_2 3}}-3^{\sqrt{\log_3 2}}\right)x-2\left(3^{\log_3 2}-2^{\log_2 3}\right)=0$, then the value of $\alpha^2+\alpha\beta+\beta^2$ is equal to :

1) 3 2) 5 3) 7 4)11

9. Let z be a complex number satisfying $\frac{1}{2}\le|z|\le 4$, then sum of greatest and least value of $\left|z+\frac{1}{z}\right|$ is :

1) 65/4 2) 65/16 3) 17/4 4) 17

10. The coefficient of x^{50} in the expansion of $\sum_{k=0}^{100} {}^{100}C_k\left(x-2\right)^{100-k}3^k$ is also equal to:

1) Number of ways in which 50 identical books can be distributed in 100 students, if each students can get atmost one book.

2) Number of ways in which 100 different white balls and 50 identical red balls can be arranged in a circle, if no two red balls are together.

3) Number of dissimilar terms in $\left(x_1+x_2+x_3+....+x_{50}\right)^{51}$.

4) Number of ways in which 50 identical books can be distributed in 100 students

11. In the polynomial function $f(x)=(x-1)\left(x^2-2\right)\left(x^3-3\right).....\left(x^{11}-11\right)$ the coefficient of x^{60} is:

1)7 2) -4 3) -7 4) 1

12. X and Y are weak students in mathematics and their chance of solving a problem correctly are 1/8 and 1/12 respectively. They are given a question and they obtain the same answer. If the probability of common mistake is $\frac{1}{1001}$, then probability that the answer was correct is a/b (a and b are coprimes). Then $|a-b|=$

1) 1 2) 11 3) 19 4) 27

13. Let $S=\left\{(x,y)/x,y\in R, x^2+y^2-10x+16=0\right\}$. The largest value of $\frac{y}{x}$ can be put in the form $\frac{m}{n}$ where m,n are relatively prime natural number, then $m^2+n^2=$

1) 16 2) 7 3) 25 4) 9

14. If locus of mid-point of any normal chord of the parabola : $y^2 = 4x$ is $x - a = \frac{b}{y^2} + \frac{y^2}{c}$; where $a, b, c \in N$, then $(a+b+c)$ equal to :

1) 5 2) 8 3) 10 4) None of these

15. A normal to the hyperbola $\frac{x^2}{4} - \frac{y^2}{1} = 1$ has equal intercepts on positive x and positive y-axes. If this normal touches the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, then $3(a^2 + b^2)$ is equal to :

1) 5 2) 25 3) 16 4) None of these

16. Let a,b and c be the 7th, 11th and 13th terms respectively of a non-constant AP. If these are also the three consecutive terms of a GP then $\frac{c}{a} =$ _______

1) $\frac{7}{13}$ 2) 2 3) 4 4) $\frac{1}{4}$

17. Let f(x) = $8x^3 + 3x$ and $f^{-1}(x)$ be the inverse of $f(x)$ then $\underset{x \to \infty}{\mathrm{lt}} \frac{x^{\frac{1}{3}}}{f^{-1}(8x) - f^{-1}(x)}$

1) 0 2) 1 3) 2 4) 4

18. Let $\mathrm{f}:[0,1] \to \mathrm{R}$ be such that $\mathrm{f}(\mathrm{xy}) = \mathrm{f}(\mathrm{x})\mathrm{f}(\mathrm{y})$, $\forall x, y \in [0,1]$ and $f(0) \neq 0$; If y = y(x) satisfies the differential equation , $\frac{dy}{dx}$ = f(x) with $\mathrm{y}(0) = 1$ then $y\left(\frac{1}{3}\right) + y\left(\frac{2}{3}\right) =$

1) 2 2) 3 3) 5 4) 4

19. The plane contains the line $\frac{x-3}{2} = \frac{y+2}{-1} = \frac{z-1}{3}$ and also containing its projection on the plane $2\mathrm{x} + 3\mathrm{y} - \mathrm{z} = 5$; contains which of the following points

1) (2,0,-2) 2) (-2,2,2) 3) (0,-2,2) 4) (2,2,0)

20. Let $P = \begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 9 & 3 & 1 \end{pmatrix}$; Q = $[q_{ij}]$ be two 3×3 matrices that such that $Q - P^5 = I_3$ then

$\frac{q_{21} + q_{31} + q_{32}}{q_{21}} =$

1) 15 2) 11 3) 135 4) 10

21. A data consists of n observations $x_1, x_2 .... x_n$. If $\sum_{i=1}^{n}(x_i+1)^2 = 9n$; and $\sum_{i=1}^{n}(x_i-1)^2 = 5n$ then the standard deviation of this data is –

1) 2 2) $\sqrt{5}$ 3) 5 4) $\sqrt{7}$

22. If [x] denotes the greatest integer ≤x then the values of $S=\left[\sqrt{1}\right]+\left[\sqrt{2}\right]+\left[\sqrt{3}\right]+............\left[\sqrt{2024}\right]$ is

1) 59000 2) 58750 3) 59730 4) 59686

23. Let f(x) = $\begin{cases} \int_0^x \{1+|1-t|\}dt & if\ x>2 \\ 5x-7 & if\ x\le 2 \end{cases}$

1) f is not continuous at x =2

2) f is continuous but not differentiable at x = 2

3) f is differential every where

4) $f^1(2^+)$ does not exist

24. Suppose A,B,C are angels of a triangle and let $\Delta = \begin{vmatrix} e^{2iA} & e^{-ic} & e^{-iB} \\ e^{-ic} & e^{2iB} & e^{-iA} \\ e^{-iB} & e^{-iA} & e^{2ic} \end{vmatrix}$ Then $\Delta =$______

1) 0 2) 4 3) -4 4) -1

25. If real numbers p,q,r satisfy $4p+3q+r=7$ Then the least value of $2p^2+q^2+r^2$ is

1) $\sqrt{7}$ 2) $\frac{7}{\sqrt{2}}$ 3) $\frac{\sqrt{7}}{2}$ 4) $\frac{49}{18}$

26. The expression $\sim(\sim p \rightarrow q)$ is logically equivalent to

1) $\sim p \wedge \sim q$ 2) $p \wedge q$ 3) $\sim p \wedge q$ 4) $p \wedge \sim q$

27. The number of solutions of tanx+tan2x+tan3x =tanx tan2x tan3x in $(0,\pi)$ is ______

1) 3 2) 4 3) 1 4) 2

28. $\sin^{-1}(\sin 10)+\cos^{-1}(\cos 10)+\tan^{-1}(\tan 10)=$ x then [x] = {where [ ] denotes GIF }

1) 1 2) 2 3) 3 4) 4

29. Let O be the origin of the coordinate system in the Cartesian plane, $\overline{OP}$ and $\overline{OR}$ be vectors making angles 45^0 and 135^0 respectively with the +ve directions of the x-axis (i.e in the counter clock wise) Rectangle OPQR is completed and M is the mid point of $\overline{PQ}$. If the line $\overline{OM}$ meets the diagonal PR at T and $\left|\overline{OP}\right|=3; \left|\overline{OR}\right|=4$ then $\overline{OT}=$________

1) $\frac{1}{2}\left(\bar{i}+\bar{j}\right)$ 2) $\frac{2}{3}\left(\bar{i}+5\bar{j}\right)$ 3) $\frac{\sqrt{2}}{3}\left(\bar{i}-5\bar{j}\right)$ 4) $\frac{\sqrt{2}}{3}\left(\bar{i}+5\bar{j}\right)$

30. If $\bar{a},\bar{b},\bar{c}$ are unit vectors satisfying $\left|\bar{a}-\bar{b}\right|^2+\left|\bar{b}-\bar{c}\right|^2+\left|\bar{c}-\bar{a}\right|^2=9$

Then $\left|2\bar{a}+7\bar{b}+7\bar{c}\right|=$____________

1) 3 2) 4 3) 5 4) 6

PHYSICS

31. The least count of a stop watch is $\frac{1}{5}s$. The time of 20 oscillations of a pendulum is measured as 25s. The percentage error in the measurement of time will be .

1)0.1% 2)0.8% 3)1.8% 4)8%

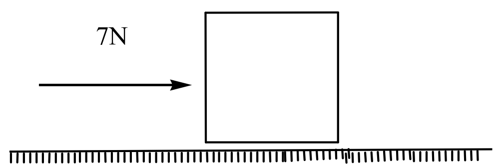
32. A stone falls from rest. The total distance covered by it in the last second of its motion is equal to the distance covered in the first three seconds. What is the height from which the stone was dropped ? take $g=10m/s^2$

1)25m 2)100m 3)125m 4)200m

33. one second after projection, a stone moves at an angle of 45^0 with the horizontal . two seconds after projection, it moves horizontally. Its angle of projection is (in degrees) ($g=10m/s^2$).

1) 60^0 2) $\tan^{-1}(4)$ 3) $\tan^{-1}(3)$ 4) $\tan^{-1}(2)$

34. A block of mass 2kg is placed on the floor ($\mu=0.4$). A horizontal force of 7N is applied on the block. The force of friction between the block and floor is

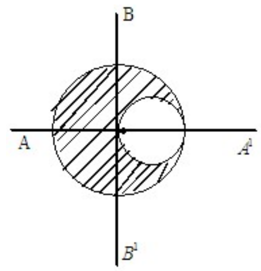


1)8N 2)7N 3)5N 4)6N

35. Particle A makes a perfectly elastic collision with another particle B at rest . They fly apart in opposite direction with equal speeds. If their masses are m_A and m_B respectively, then

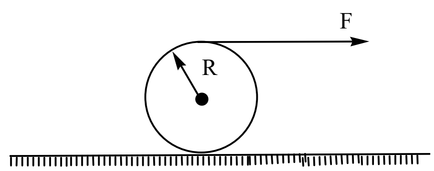
1) $2m_A = m_B$ 2) $3m_A = m_B$ 3) $4m_A = m_B$ 4) $\sqrt{3}m_A = m_B$

36. From a uniform sphere of mass M and radius R , a cavity of diameter R is created as shown. Find the ratio of moment of inertia of the sphere left about AA^1 and BB^1 .



1) $\frac{15}{28}$ 2) $\frac{28}{15}$ 3) $\frac{31}{30}$ 4) $\frac{62}{57}$

37. A ring of mass m and radius R is acted upon by a force F as shown in the figure . There is sufficient friction between the ring and the ground. The force of friction necessary for pure rolling is



1) $\frac{F}{2}$ *forward* 2) $\frac{F}{3}$ *forward* 3) zero 4) $\frac{F}{4}$ *back ward*

38. Given that mass of the earth is M and its radius is R .A body is dropped from a height equal to the radius of the earth above the surface of the earth .when it reaches the ground , its speed will be .

1) $\frac{GM}{R}$ 2) $\sqrt{\frac{GM}{R}}$ 3) $\sqrt{\frac{2GM}{R}}$ 4) $\frac{2GM}{R}$

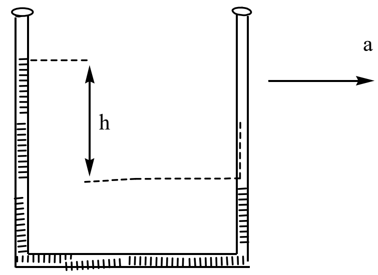
39. Amplitude of vibration is $A = \frac{F_0}{p-q+r}$. Resonance will occur when

1) $p = 0, q = r$ 2) $p = q = r$ 3) $p = -r, q = 0$ 4) both(1)&(3)

40. A 1000 kg lift is tied with metallic wires of maximum safe stress of $1.4\times10^8 N/m^2$ .If the maximum acceleration of the lift is $1.2 m/s^2$, then the minimum diameter of the wire is.

1)0.01m 2)0.01cm 3)0.001m 4)0.02cm

41. When at rest a liquid stands at the same level in the tubes shown in figure. But as indicated a height difference h occurs when the system is given an acceleration 'a' towards the right. Here 'h' is equal to



1) $\frac{al}{2g}$ 2) $\frac{gL}{2a}$ 3) $\frac{gL}{a}$ 4) $\frac{aL}{g}$

42. For an enclosure maintained at 2000K, the maximum radiation occurs at wavelength λ_m. If the temperature is raised to 3000K , the peak will shift to

1)0.5 λ_m 2) λ_m 3) $\frac{2}{3}\lambda_m$ 4) $\frac{3}{2}\lambda_m$

43. 10gm of ice at $-20^0 C$ is kept into a calorimeter containing 10gm of water at $10^0 C$. The specific heat of water is twice that of ice .When equilibrium is reached, the calorimeter will contain

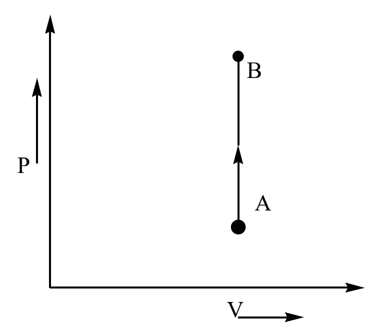
1)20gm of water

2)20gm of ice

3)10gm of ice and 10 gm of water

4)5 gm of ice and 15 gm of water

44. The following graph represents



1) $\Delta p = 0$ 2) $\Delta W = 0$ 3) $\Delta T = 0$ 4) $\Delta Q = 0$

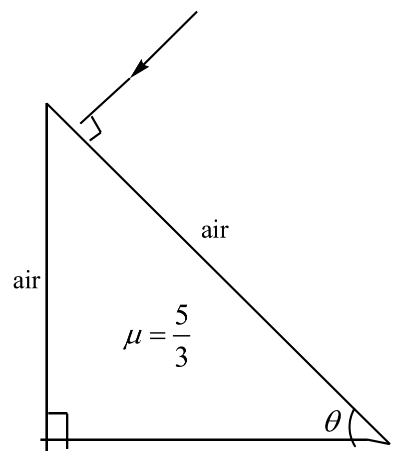
45. 2moles of a diatomic gas is enclosed in a vessel. When a certain amount of heat is supplied, 50% of the gas molecules get dissociated, but there is no rise in temperature. What is the heat supplied if temperature is T ?

1) RT 2) $\frac{RT}{2}$ 3) $\frac{11}{2}RT$ 4) $5RT$

46. If the first overtone of a closed pipe of length 50cm has the same frequency as the first overtone of an open pipe, then the length of the open pipe is.

1)100cm 2)200cm 3)66.6cm 4)33.3cm

47. A ray of light makes normal incidence on the diagonal face of a right angled prism as shown in figure .If $\theta = 37^0$,then the angle of deviation is ($\sin 37^0 = \frac{3}{5}$)



1) 53^0 2) 127^0 3) 106^0 4) 90^0

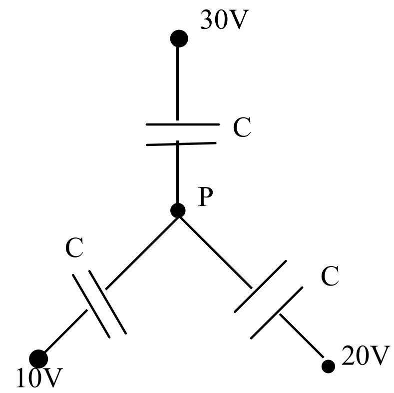
48. In young's double slit experiment, the intensity at a point where path difference is $\frac{\lambda}{6}$ is I. If I_0 denotes the maximum intensity, then $\frac{I}{I_0}$ is

1) $\frac{1}{\sqrt{2}}$ 2) $\frac{\sqrt{3}}{2}$ 3) $\frac{1}{2}$ 4) $\frac{3}{4}$

49. A uniform electric field $\vec{E} = E_0\left(-\hat{j}\right)$ exists in space. An electron is given velocity $\vec{V} = V_0\hat{i}$ from origin. If the equation of path of electron is y= $y = \left(\frac{eE_0}{2m}\right)\frac{x^a}{V_0^{\,b}}$. Then

1) $a = 2, b = 2$ 2) $a = 2, b = 1$ 3) $a = 2, b = -1$ 4) $a = 1, b = -2$

50. Find the potential of point 'p'



1) 15V 2) 60v 3) 10V 4) 20V

51. The drift velocity of free electron in a metal wire of a given potential gradient along it is V_d .If this potential gradient is doubled, the new drift velocity will be

1) V_d 2) $2V_d$ 3) $\frac{V_d}{2}$ 4) $4V_d$

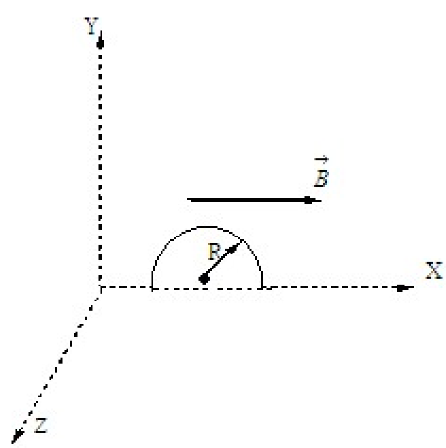
52. A changed particle , having charge 'q' accelerated through a potential difference 'V' enters a perpendicular magnetic field in which it experiences a force F . If 'V' is increased to 5V , the particle will experience a force

1) F 2) 5F 3) $\frac{F}{5}$ 4) $\sqrt{5}F$

53. Relative permeability of iron is 5500. Its magnetic susceptibility is

1) 5500×10^{7} 2) 5499 3) 5501 4) 5500×10^{-7}

54. A semicircle conducting ring of radius R is placed in XY plane as shown in figure. A uniform magnetic field exist along x-axis .No emf will be induced if



1) It moves along x-axis 2) it moves along y-axis

3) it movers along z-axis 4) all of these.

55. An alternating current of 1.5 mA with angular frequency 100 rad/s flows through a $10k\Omega$ resistor and a 0 $0.50\,\mu F$ capacitor in series. The rms potential drop across the capacitor is

1)4.8V 2)30V 3)15V 4)190V

56. An electromagnetic wave travels in a medium with speed $2\times10^{8}\,m/s$. If relative permeability of medium is 1.0 , then relative permittivity is.

1)2.25 2)4.5 3)0.5 4)4

57. If longest wavelength of balmer series of H atom is λ then shortest wavelength of lyman series will be

1) $\frac{5}{36}\lambda$ 2) $\frac{5}{9}\lambda$ 3) $\frac{36}{5}\lambda$ 4) $\frac{9}{5}\lambda$

58. If the half life of a radio active is T, then the fraction that would remain after a time $\frac{T}{2}$ is

1) $\frac{1}{\sqrt{2}}$ 2) $\frac{\sqrt{2}-1}{\sqrt{2}}$ 3) $\frac{1}{2}$ 4) $\frac{3}{4}$

59. A transistor is connected in common base configuration, the collector supply is 8V and the voltage drop across a resistor of 800Ω in the collector circuit is 0.5V . If the current gain is 0.96, then the base current is.

1) 50×10^{-6} A 2) 450×10^{-6} A

3) 80×10^{-6} A 4) 26×10^{-6} A

60. If a carrier wave of 1000 KHz in used to carry the message signal, the length of transmitting antenna will be equal to

1)3m 2)30m 3)75m 4)3000m

CHEMISTRY

61. 14g of an element, X combines with 16 g of oxygen. On the basis of this information, which of the following is a correct statement ?

1)The element X could have an atomic mass of 7 and its oxide is XO

2)The element X could have an atomic mass of 14 and its oxide formula is X_2O

3)The element X could have an atomic mass of 7 and its oxide is X_2O

4)The element X could have an atomic mass of 14 and its oxide is XO_2

62. The angular momentum in Bohr's H-atom is proportional to :

1) r^2 2) $\frac{1}{r}$ 3) $\sqrt{r}$ 4) $\frac{1}{\sqrt{r}}$

63. CO_2 has the same geometry as :

(1) $HgCl_2$ **(2)** NO_2 **(3)** $SnCl_4$ **(4)** C_2H_2

1)1 and 3 2)2 and 4 3)1 and 4 4)3 and 4

64. Dipole moment of HCl molecule is found to be 0.816D. Assuming HCl bond length to be equal to $1\overset{0}{A}$, the % ionic character of HCl molecule is:

1)10% 2)17% 3)27% 4)37%

65. The ratio of average speed of an oxygen molecule to rms speed of a nitrogen molecule at the same temperature is :

1) $\left(\frac{3\pi}{7}\right)^{1/2}$ 2) $\left(\frac{7}{3\pi}\right)^{1/2}$ 3) $\left(\frac{3}{7\pi}\right)^{1/2}$ 4) $\left(\frac{7\pi}{3}\right)^{1/2}$

66. Which are correct statements?

1) H_2O_2 is less associated liquid than H_2O

2) H_2O_2 is paramagnetic in nature

3) H_2O_2 acts as oxidant and reductant both

4) H_2O_2 reacts with $K_2Cr_2O_7$ in acidic medium to give blue color solution

1)3,4 2)1,3,4 3)2,3,4 4)1,2,3

67. 0.75g chloroplatinate chloride of a mono-acid base on ignition gave 0.245 g platinum. The molar mass of the base is :

1)75.0 2)93.5 3)100 4)80.0

68. The Gibbs energy for the decomposition of Al_2O_3 at 500^0C is as follows:

$\frac{2}{3}Al_2O_3 \rightarrow \frac{4}{3}Al + O_2; \Delta_r G = 966 kJmol^{-1}$. The potential difference needed for electrolytic reduction of Al_2O_3 at $500^0 C$ is at least:

1)5.0 V 2)4.5 V 3)3.0 V 4)2.5 V

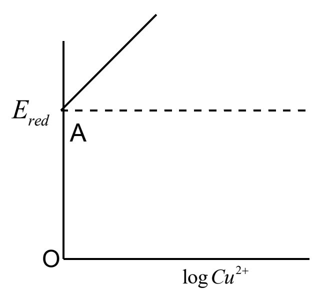
69. $CsBr$ crystallizes in a body centered cubic lattice. The unit cell length is 436.6 pm. Given that the atomic mass of $Cs = 133$ and that of $Br = 80$ amu and Avogadro number being $6.02 \times 10^{23} mol^{-1}$ the density of CsBr is :

1) $8.25\, g/cm^3$ 2) $4.25\, g/cm^3$ 3) $42.5\, g/cm^3$ 4) $0.425\, g/cm^3$

70. The degree of dissociation (α) of a weak electrolyte, A_xB_y is related to Vant't Hoff factor (i) by the expression:

1) $\alpha = \frac{i-1}{(x+y-1)}$ 2) $\alpha = \frac{i-1}{(x+y+1)}$ 3) $\alpha = \frac{x+y-1}{i-1}$ 4) $\alpha = \frac{x+y+1}{i-1}$

71. $Cu^{2+} + 2e \rightarrow Cu; \log\left[Cu^{2+}\right] vs, E_{red}$ graph is of the type as shown in figure where $OA = 0.34V$, then $E_{Cu^{2+}(0.1M)|Cu(s)}$ will be:



1) $0.34 + \frac{0.0591}{2}V$ 2) $0.34 + 0.0591V$ 3) $0.34V$ 4)None of these

72. For a certain reaction of order 'n' the time for half change $t_{1/2}$ is given by :

$t_{1/2} = \frac{2-\sqrt{2}}{K} \times c_0^{1/2}$, where K is rate constant and c_0 is initial concentration. The value of n is :

1)1 2)2 3)0 4)0.5

73. BCl_3 is a planar molecule where as NCl_3 is pyramidal because:

1) BCl_3 has no lone pair of electrons but NCl_3 has lone pair of electrons

2) $B-Cl$ bond is more polar than $N-Cl$ bond

3) nitrogen atom is smaller than boron atom

4) $N-Cl$ bond is more covalent than $B-Cl$ bond

74. In the reaction $2KI+H_2O+O_3 \rightarrow 2KOH+O_2+A$ the compound A is :

1) I_2 2) I_2O_5 3) O_3 4) N_2O

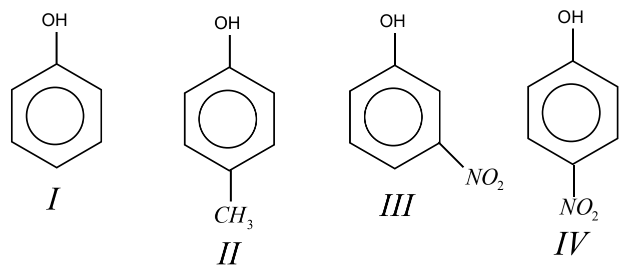
75. Acetaldoxime reacts with P_2O_5 to give:

1) CH_3CN 2) C_2H_5CNO 3) C_2H_5CN 4) All of these

76. The best coagulant for the precipitation of $Fe(OH)_3$ is:

1) Na_2HPO_3 2) $NaNO_3$ 3) Na_3PO_4 4) Na_2SO_4

77. In the following compounds, the order of acidity is :



1)II > IV > I > II 2)I > IV > III > II

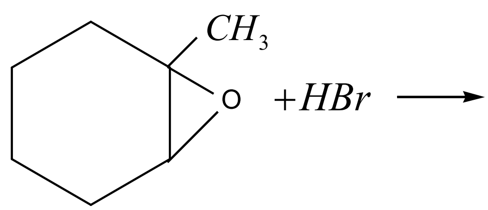
3)II > I > III >IV 4)IV >III > I >II

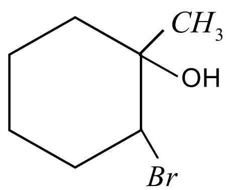
78. The product in the reaction is :

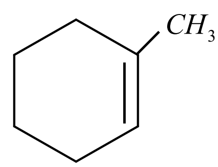
$$C_2H_5OH \xrightarrow{P+I_2} A \xrightarrow[Ether]{Mg} B \xrightarrow{HCHO} C \xrightarrow{H_2O} D$$

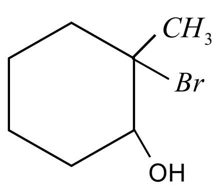
1)Propanal 2)butanal 3)n-butanol 4)n-propanol

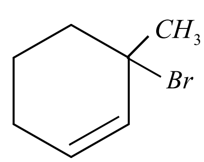
79. The product of the reaction



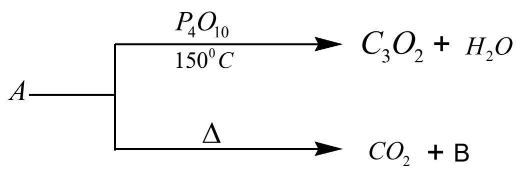
1) 

2) 

3) 

4) 

80. In the following reactions



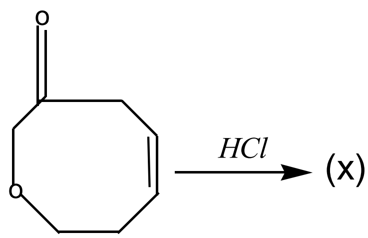
The compound (s) (A) and (B) respectively are

1) $CH_3CH_2COOH\ and\ C_2H_6$

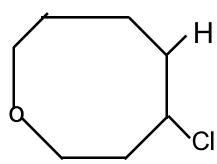
2) $CH_2(COOH)_2\ and\ CH_4$

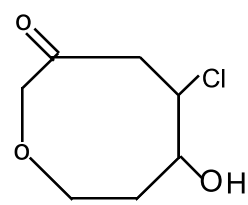
3) $CH_3CH(COOH)_2\ and\ CH_3CH_2COOH$

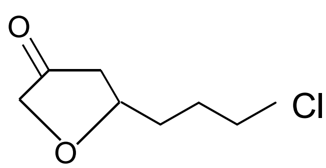
4) $CH_2(COOH)_2\ and\ CH_3COOH$

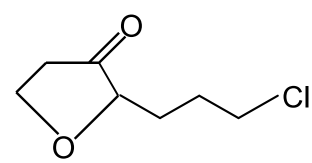
81. 

X is

1) 

2) 

3) 

4) 

82. $2X + H_2SO_4 + H_2O_2 \rightarrow 2Y + 2K_2SO_4 + 2H_2O$

$2Y + 2KOH + H_2O_2 \rightarrow 2X + 2H_2O + O_2$

The incorrect statement about compounds X and Y in the above reactions is

1) Oxidation number of central metal ion in X is +3 and Y is +2

2) H_2O_2 acts as oxidizing agent in reaction with X and reducing agent in reaction with Y

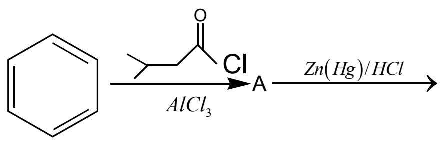
3) Oxidation number of central metal Ion in X is +2 and in Y is +3

4) Both X and Y are iron complexes

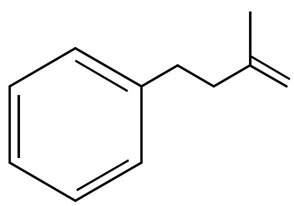
83. For the reaction: $2FeS_2 + \frac{11}{2}O_2 \rightarrow Fe_2O_3 + 4SO_2$. What will be the equivalent weight of FeS_2 , if the molecular weight of FeS_2 is M ?

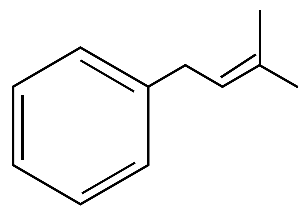
1) $M/8$ 2) M 3) $M/11$ 4)Can't be calculated

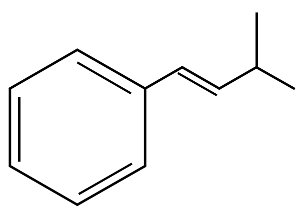
84. The major product (D) in the following reaction sequence is

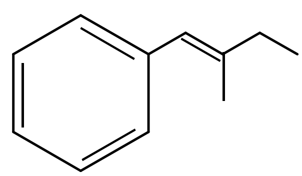


$B \xrightarrow{Br_2/hv} C \xrightarrow{(CH_3)_3COK} D$ Major

1) 

2) 

3) 

4) 

85. Which of the following compound is formed when XeF_4 react with water ?

1) XeO_3 2) XeO_4 3) $XeOF_4$ 4) $XeOF_2$

86. Identify the statement which is not correct regarding $CuSO_4$

1)It reacts with KI to give iodine

2)It reacts with $NaOH$ and glucose to give Cu_2O

3)It reacts with KCl to give Cu_2Cl_2

4)It gives CuO on heating

87. $SO_3^{-2} + S^{-2} + I_2$ (Calculated amount) $\rightarrow X + I^{-1}$. Then "X" in this reaction is

1) SO_4^{-2} 2) IO_3^{-1} 3) $S_2O_3^{-2}$ 4) IO_4^{-1}

88. The reduction of nitric oxide gas by hydrogen gas as represented below:

$2NO+2H_2 \rightarrow N_2+2H_2O$

Takes place in the sequence of following steps:

$I.NO+NO \rightleftharpoons N_2O_2$ (fast and reversible)

$II.N_2O_2+H_2 \rightleftharpoons N_2O+H_2O$ (slow)

$III.N_2O+H_2 \rightarrow N_2+H_2O$ (fast)

The reaction follows the rate law $+\frac{d[N_2]}{dt}=k[NO]^2[H_2]$

If $k=6.0\times10^{-4}mole^{-2}L^2\min^{-1}$ and k_2 the rate constant of Step II be $3.0\times10^{-2}mole^{-1}L\min^{-1}$, the numerical value of the equilibrium constant (k_c) for the Step I is :

1) 1.8×10^{-8} 2) 2×10^{-2} 3) 5×10^{-7} 4)None of these

89. An element has successive ionization enthalpies as 940(first), 2080, 3090, 4140, 7030, 7870,16000 and $19500 kJ\,mol^{-1}$. To which group of the periodic table does this element belong ?

1)14 2)15 3)16 4)17

90. Which of the following reagent can be used to identify Fe^{+3} in presence of Fe^{+2}

1) $NaOH$ 2) NH_4OH 3) $KSCN$ 4) All of these

KEY SHEET

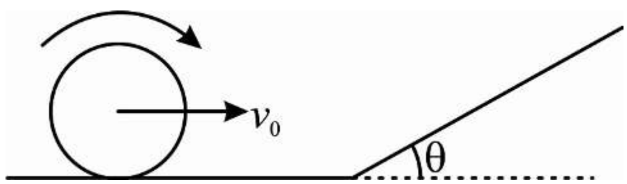
1	**2**	2	**1**	3	**2**	4	**3**	5	**4**
6	**3**	7	**4**	8	**3**	9	**3**	10	**1**
11	**4**	12	**1**	13	**3**	14	**2**	15	**2**
16	**4**	17	**3**	18	**2**	19	**1**	20	**2**
21	**2**	22	**3**	23	**2**	24	**3**	25	**4**
26	**1**	27	**4**	28	**1**	29	**4**	30	**3**
31	**2**	32	**3**	33	**4**	34	**2**	35	**2**
36	**4**	37	**3**	38	**2**	39	**4**	40	**1**
41	**4**	42	**3**	43	**3**	44	**2**	45	**2**
46	**3**	47	**2**	48	**4**	49	**1**	50	**4**
51	**2**	52	**4**	53	**2**	54	**4**	55	**2**
56	**1**	57	**1**	58	**1**	59	**4**	60	**3**
61	**3**	62	**3**	63	**3**	64	**2**	65	**2**
66	**1**	67	**2**	68	**4**	69	**2**	70	**1**
71	**1**	72	**1**	73	**4**	74	**1**	75	**1**
76	**3**	77	**4**	78	**4**	79	**3**	80	**4**
81	**3**	82	**1**	83	**3**	84	**3**	85	**1**
86	**3**	87	**3**	88	**2**	89	**3**	90	**3**

Sec: Sr. IIT **Date: 21-02-24**

Daily Practices Sheet - 03

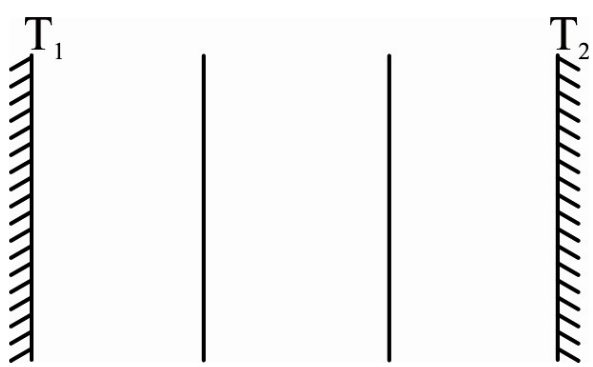
PHYSICS

1. In a model experiment, a rigid surface consists of a rough horizontal plane and an inclined plane connecting to it. A thin hoop of radius r is rolling towards the slope without slipping at a velocity v_0, perpendicular to the base of the slope. The slope connects to the horizontal plane with a smooth curve of radius R > r, which is considered part of the slope, such that the hoop will move up the slope without collision. In three different models of the experiment, everything is kept same except the friction coefficient of slope which is $\mu_1 = 0, \mu_2 = \tan\theta, \mu_3 = 2\tan\theta$ respectively. In each case the height to which hoop rises on the incline before stopping momentarily is $\Delta h_1, \Delta h_2, \Delta h_3$ respectively. Then



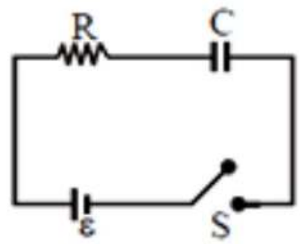
1) $\Delta h_1 < \Delta h_2 < \Delta h_3$ 2) $\Delta h_1 > \Delta h_2 > \Delta h_3$ 3) $2\Delta h_1 = \Delta h_2$ 4) $2\Delta h_2 = \Delta h_3$

2. Two large black plane surfaces are maintained at constant temperatures T_1 and T_2 ($T_1 > T_2$). Two thin black plates are placed between the two surfaces and in parallel to these. After some time, steady conditions are obtained. By what factor (η) is the steady heat flow reduced due to the presence of black plates?



1) $\eta = \frac{1}{2}$ 2) $\eta = \frac{1}{3}$ 3) $\eta = 1$ 4) $\eta = \frac{1}{4}$

3. If switch S is closed at t = 0 then the time at which power supplied by battery is equal to rate of energy storage in capacitor is



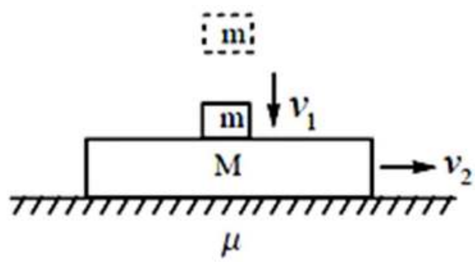
1) t = 0

2) t = 4RC

3) t = 5RC

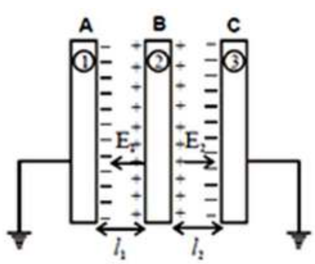
4) It never happens (except $t \to \infty$) because resistor always consumes energy

4. A block of mass M = 5kg is moving on a horizontal plane. An object of mass m = 1kg is dropped onto the block, hitting it with a vertical velocity of $v_1 = 10\,m/s$. The speed of the block at the same instant is $v_2 = 2\,m/s$ on the floor. The collision is momentary and object sticks to the block upon collision. What will be the speed of the block just after the collision if the coefficient of friction between the block and the horizontal plane is $\mu = 0.4$?



1) 0.8 m/s 2) 1 m/s 3) 1.4 m/s 4) 1.67 m/s

5. Suppose Electric field between plate (A) and (B) is E_1 and between (B) and (C) is E_2 then



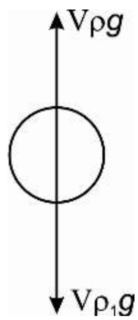
1) $\frac{E_1}{E_2} = \frac{\ell_1}{\ell_2}$ 2) $\frac{E_1}{E_2} = \frac{\ell_2}{\ell_1}$ 3) $E_1\ell_1 > E_2\ell_2$ 4) none of these

6. If the cathode of a photocell is illuminated with a light of increasing frequency, the anode current will start at a frequency of $3\times10^{14}Hz$. A capacitor with capacitance 1pF is connected between the anode and the cathode of this photocell and the cathode is illuminated with light of frequency $7\times10^{14}Hz$. Assuming the illumination is long enough, find the approximate number of electrons arriving on the anode?

$\left(h=6.63\times10^{-34}Js, e=1.6\times10^{-19}C\right)$

1) 1×10^{10} 2) 2×10^{12} 3) 3×10^{15} 4) 4×10^{17}

7. The density ρ of a liquid varies with depth h from the free surface as $\rho = \text{kh}$. A small body of density ρ_1 is released from the surface of liquid. The body will



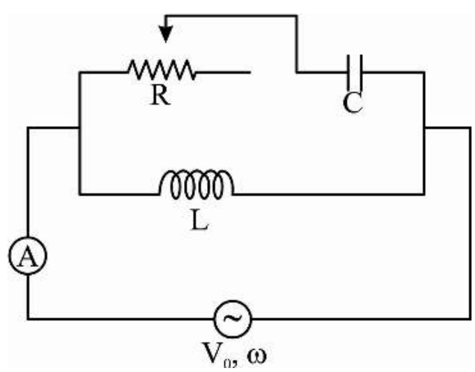
1) come to a momentary rest at a depth $\frac{2\rho_1}{\text{k}}$ from the free surface

2) execute simple harmonic motion about a pint at a depth $\frac{\rho_1}{\text{k}}$ from the surface

3) execute simple harmonic motion of amplitude $\frac{\rho_1}{\text{k}}$

4) all of the above

8. In the shown 'AC' circuit L, C and V_0 are fixed. What is the value of ω such that reading of hot wire ammeter shown is independent of 'R'?



1) $\frac{1}{\sqrt{LC}}$ 2) $\sqrt{\frac{2}{LC}}$ 3) $\sqrt{\frac{3}{LC}}$ 4) $\frac{2}{\sqrt{LC}}$

9. At the center of a non-uniform ring of radius R, made up of two uniform halves of mass 2M and M (G: Universal gravitation constant)

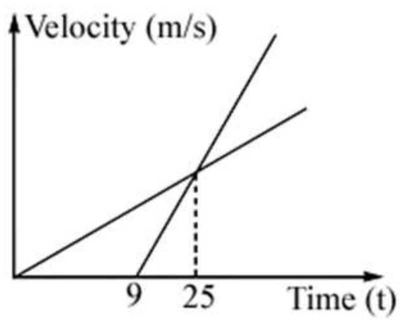
1) field and potential both are zero

2) field is zero but potential is $\frac{-3GM}{R}$

3) field is zero but potential is $\frac{-GM}{R}$

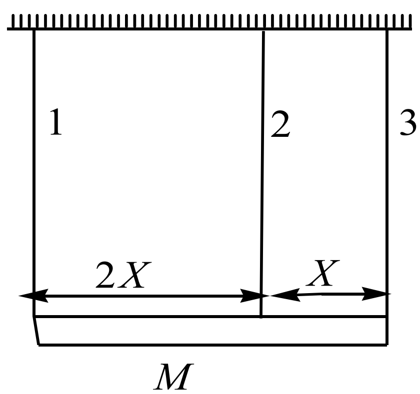
4) magnitude of field is $\frac{2GM}{\pi R^2}$ and potential is $-\frac{3GM}{R}$

10. A material particle is chasing the other one and both of them are moving on the same straight line. Their motion after they pass a particular point is recorded and the data obtained is shown by velocity-time graphs of the particles. How long after the start will the chase end?



1) 16s 2) 25s 3)45s 4) insufficient information

11. Three vertical wires of same length 1,2and 3 are supporting a rod mass M in horizontal position in final equilibrium as shown in the figure. The wires are having equal cross-sectional area. It is given that $Y_2 = Y_3$. The wires 1 and 3 are attached at the extreme ends of the rod Choose the correct option.



1) $T_1 = 2T_3$ 2) $2T_1 = T_3$ 3) $T_2 = T_3$ 4) $T_1 = \frac{Mg}{5}$

12. A rigid square loop is dipped in a wetting liquid and is taken out. A loop of light inextensible thread of total length 5 units is gently put on the liquid film formed in the square frame. Now a hole is pricked inside the thread loop. Take the length of side of square frame is 4 units. Surface tension of liquid is S. Neglect any friction between square frame and thread. What will be final shape of thread after the hole is pricked?

1) a square whose centre must lie at centre of square frame

2) a circle whose centre must lie at centre of square frame

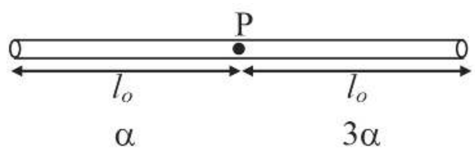
3) a circle whose centre may not lie at centre of square frame

4) a circle which must touch the edges of square frame

13. Velocity of a particle at any instant is given by the equation $\vec{v}=\left(2t\hat{i}+3t^2\hat{j}\right)$ m/s and radius of the curvature of the path is 2m. Centripetal acceleration of the particle at $t = 2s$ is

1) 80 m/s^2 2) 160 m/s^2 3) 40 m/s^2 4) 100 m/s^2

14. Two rods of same mass and same length l_o but having different coefficient of expansions α and 3α are joined at P (see figure). System is placed on frictionless surface. If temperature of whole system is changed by ΔT, the displacement of junction point is



1) $l_o\alpha\Delta T$ 2) $2l_o\alpha\Delta T$ 3) $\frac{l_o\alpha\Delta T}{2}$ 4) $\frac{3}{4}l_o\alpha\Delta T$

15. Shown here are the zero-error position (fig 1) and the measurement (fig 2). Find the actual size of the object measured? Assume least count is 0.01 cm. The units of the main scale are in millimeters. (The top scale represents main scale and the bottom Vernier scale, in each figure)

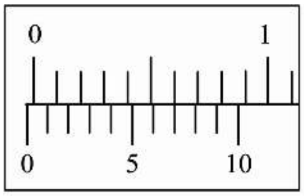


fig 1

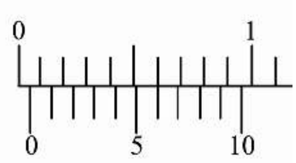


fig 2

1) 0.06 cm 2) 0.03 cm 3) 0.09 cm 4) 0.02 cm

16. Suppose that a material emits X-rays of wavelengths: $\lambda_{K_\alpha}, \lambda_{K_\beta}, \lambda_{L_\alpha}$ when it is excited by fast moving electrons; the wavelengths corresponding to $K_\alpha, K_\beta, L_\alpha$ X-rays of the material respectively. Then we can write:

1) $\lambda_{K_\beta} = \lambda_{K_\alpha} + \lambda_{L_\alpha}$
2) $\sqrt{\lambda_{K_\beta}} = \sqrt{\lambda_{K_\alpha}} + \sqrt{\lambda_{L_\alpha}}$
3) $\frac{1}{\lambda_{K_\beta}} = \frac{1}{\lambda_{K_\alpha}} + \frac{1}{\lambda_{L_\alpha}}$
4) $\frac{1}{\sqrt{\lambda_{K_\beta}}} = \frac{1}{\sqrt{\lambda_{K_\alpha}}} + \frac{1}{\sqrt{\lambda_{L_\alpha}}}$

17. An excited hydrogen atom in the state n (principal quantum number of electron: n) is moving with a velocity v (v<<c, the velocity of light in vacuum). If this atom emit a photon of frequency v opposite to the direction of its motion, while making a transition to the ground state, then(Where v_0 is the frequency of radiation emitted when the atom is at rest.)

1) $v = v_0 + \left(1 + \frac{\mathrm{v}}{\mathrm{c}}\right)$
2) $v = v_0 + \left(1 - \frac{\mathrm{v}}{\mathrm{c}}\right)$
3) $v = v_0 + \left(1 - \frac{\mathrm{v}^2}{\mathrm{c}^2}\right)$
4) $v = v_0$, independent of v

18. The objective lens of a compound microscope produces magnification of 10. In order to get an overall magnification of 100 when image is formed at 25 cm from the eye, the focal length of the eye lens should be

1) 4 cm
2) 10 cm
3) $\frac{25}{9}$ cm
4) 9 cm

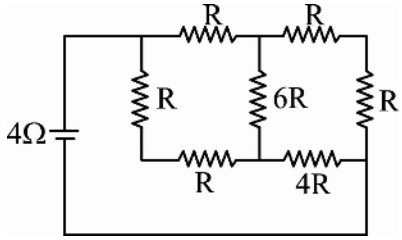
19. In a Young's double slit experiment, 12 fringes are observed to be formed in a certain segment of the screen when light of wavelength 600nm is used. If the wavelength of light is changed to 400nm, number of fringes observed in the same segment of the screen is given by

1) 12
2) 18
3) 24
4) 30

20. A stable species C can be obtained through unstable nuclides A and B. The half life period of conversion from A to C is T and that for conversion from B to C is 2T. Initially a sample contains N nuclides of A, N nuclides of B and no nuclides of C. After how much time will the nuclides of species C be equal to (27 / 16)N

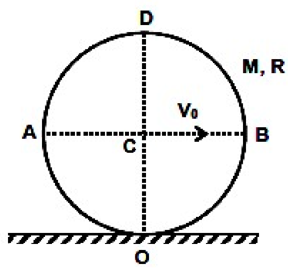
1) 2T
2) 3T
3) 4T
4) 5T

21. A battery of internal resistance 4Ω is connected to the network of resistances. In order that maximum power can be delivered to the network, the value of R in ohm should be



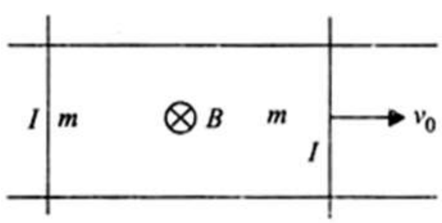
1) $\frac{4}{9}$ 2) 2 3) $\frac{8}{3}$ 4) 18

22. A uniform ring of mass m and radius R is doing uniform pure rolling motion on a horizontal surface. The velocity of the centre of the ring is V_0. The kinetic energy of the segment AOB is



1) $\frac{mV_0^2}{2}-\frac{mV_0^2}{\pi}$ 2) $\frac{mV_0^2}{2}+\frac{mV_0^2}{\pi}$ 3) $\frac{mV_0^2}{2}$ 4) mV_0^2

23. Consider parallel conducting rails separated by a distance l. There exists uniform magnetic field B perpendicular to the plane of the rails shown fig. Two conducting wires each of length l are placed so as to slide on parallel conducting rails. One of the wires is given a velocity v_0 parallel to the rails. Till steady state is achieved, loss in kinetic energy of the system is



1) zero 2) $\frac{3}{4}mv_0^2$ 3) $\frac{1}{4}mv_0^2$ 4) $\frac{3}{8}mv_0^2$

24. The displacement function of a oscillating body is giving by $x = 0.3\sin\left(10\pi t + \frac{\pi}{6}\right)$ where x and t are measured in meter and second respectively

I) The period of oscillation is 1/5s

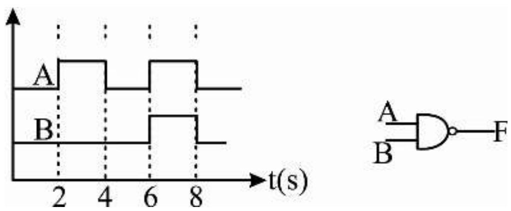
II) The body starts its motion from the equilibrium

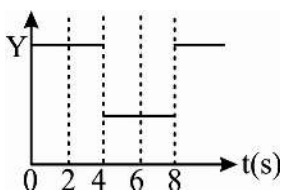
III) The minimum time the body takes to get same velocity as t=0 is 1/6s

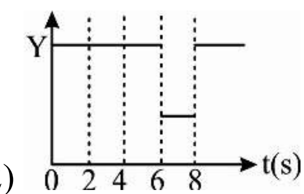
IV) Maximum velocity of the body is $3\pi m/\sec$

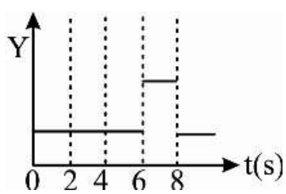
1) I,II,III,IV are correct
2) Only I,III and IV correct
3) Only I,II,III correct
4) Only I and IV correct

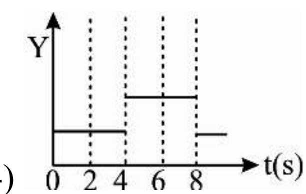
25. The real time variation of input signal A and B are as shown below. If the inputs are fed into NAND gate, then select the output signal from the following:



1) 

2) 

3) 

4) 

26. A uniform rope of mass 0.1 kg and length 2.45 m hangs from the ceiling. Time taken by a transverse wave to travel the full length of the rope is (g = 9.8 m/s^2)

1) 0.5 s
2) 1.0 s
3) 1.5 s
4) cannot be calculated until a mass is suspended at lower end of string

27. A radar wave (speed $3\times10^{8} m/s$) has a frequency $7.9\times10^{9} Hz$, is reflected from an aeroplane, shows a frequency difference of $2.7\times10^{3} Hz$ on the higher side (if we assume that Doppler's effect can be applied on light wave similar as sound wave)

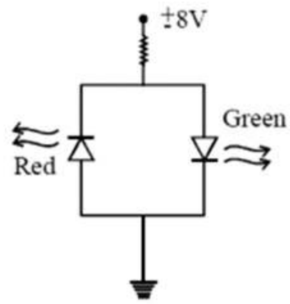
1) velocity of aeroplane is $1.87\times10^{3} km/m$

2) velocity of aeroplane in the line of sight is $1.87\times10^{3} km/m$

3) velocity of aeroplane is $1.35\times10^{3} km/m$

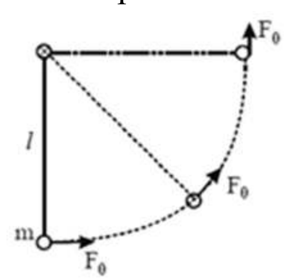
4) velocity of aeroplane in the line of sight is $1.35\times10^{3} km/m$

28. If the figure shown are 2 LED's that can be used as a polarity detector. Apply a positive source voltage and a green light results. Negative supplies result in a red light. Packages of such combination are commercially available. Find resistor R to ensure a current of 20 mA through the ON diode for the configuration. Both diodes have reverse breakdown voltage of 3V and average turn on voltage of 2V.



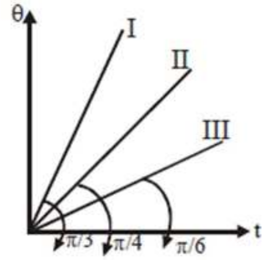
1) 250 Ω 2) 300 Ω 3) 325 Ω 4) 400 Ω

29. An agent applies force of constant magnitude F_0 always in the tangential direction as shown in the figure. Find the speed of the bob when string becomes horizontal, assuming that it is at rest at its lowest point.



1) $\sqrt{\frac{\ell}{m}(\pi F_0-2mg)}$ 2) $\sqrt{\ell g}$ 3) $\sqrt{\frac{\ell}{m}(\pi F_0-4mg)}$ 4) $\sqrt{\frac{\ell}{m}F_0}$

30. Three bodies A, B and C of masses m, m and $\sqrt{3}m$ respectively are supplied heat at a constant rate. The change in temperature θ versus time t graph for A, B and C are shown by I, II and III respectively. If their specific heat capacities are S_A, S_B, S_C respectively then which of the following relation is correct? (Initial temperature of body is 0^0C)



1) $S_A > S_B > S_C$ 2) $S_B = S_C < S_A$ 3) $S_A = S_B = S_C$ 4) $S_B = S_C > S_A$

MATHS

31. The points represented by $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ are coplanar and $(\sin A)\vec{a} + (2\sin 2B)\vec{b} + (3\sin 3C)\vec{c} - 4\vec{d} = \vec{0}$ then, the least value of $\frac{21}{8}(\sin^2 A + \sin^2 2B + \sin^2 3C)$ is

1) 1 2) 2 3) 4 4) 3

32. A and B are two non-singular square matrices of each 3 x 3 such that AB = A and BA = B and $|A+B| \neq 0$ then

1) $|A+B| = 0$ 2) $|A+B| = 8$ 3) $|A-B| = 1$ 4) $|A+B| = 2$

33. For integer n > 1, the digit at units place in the number $\sum_{r=0}^{20} r! + 2^{2^n}$ is

1) 0 2) 4 3 6 4) 9

34. If $\sum_{i=1}^{4}(x_i^2 + y_i^2) \le 2x_1x_3 + 2x_2x_4 + 2y_2y_3 + 2y_1y_4$ then points $(x_1, y_1), (x_2, y_2), (x_3, y_3), (x_4, y_4)$ are

1) vertices of a rectangle 2) collinear
3) concyclic 4) none of these

35. If $(a\cos\theta_1, a\sin\theta_1), (a\cos\theta_2, a\sin\theta_2), (a\cos\theta_3, a\sin\theta_3)$ represent the vertices of an equilateral triangle inscribed in a circle, then

1) $\sum \sec\theta_i = 0$ 2) $\sum \sin\theta_i = 0$ 3) $\sum \tan\theta_i = 0$ 4) $\sum \cot\theta_i = 0$

36. If $y = x\log\left(\frac{x}{a+bx}\right)$ then $x^3\frac{d^2y}{dx^2}$ is equal to

1) $\left(x\frac{dy}{dx} - y\right)^2$ 2) $\frac{a^2x}{(a+bx)^2}$ 3) $\left(\frac{dy}{dx} - y\right)^2$ 4) $\left(x\frac{dy}{dx} + y\right)^2$

37. If the line $x+2y+4=0$, cutting the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ in points whose eccentric angles are 30^0 and 60^0 subtends right angle at the origin then its equation is

1) $\frac{x^2}{8}+\frac{y^2}{4}=1$ 2) $\frac{x^2}{16}+\frac{y^2}{4}=1$ 3) $\frac{x^2}{4}+\frac{y^2}{16}=1$ 4) none of these

38. If the two roots of the equation $(c-1)(x^2+x+1)^2-(c+1)(x^4+x^2+1)=0$ are real and distinct and $f(x)=\frac{1-x}{1+x}$ then $f(f(x))+f\left(f\left(\frac{1}{x}\right)\right)$ is

1) -c 2) c 3) 2c 4) -2c

39. If a complex number z satisfies $|z-8-4i|+|z-14-4i|=10$, then the maximum value of arg(z) = $\tan^{-1}\left(\frac{11}{3\alpha}\right)$, where α =

1) 6 2) 5 3) 7 4) 4

40. If α and β are the roots of the equation $\tan x=2x$; then the value of $\int_{-1}^{1}\sin(\alpha x)\sin(\beta x)dx\ (\alpha,\beta>0)$ is

1) 0 2) 1 3) 2 4) 3

41. Let $A=\{1,2,3,4,5\}$. Then the number of one – one onto functions from A to A in which at least three elements have self-image, is equal to

1) 10 2) 11 3) 120 4) 5^5-5!

42. Let f and g be two function with f'' and g'' existing everywhere. If $f(x)g(x)=2\ \forall x$ and $f'(x).g'(x)\neq 0$. Then $\frac{f''(x)}{f'(x)}=\frac{g''(x)}{g'(x)}$

1) has at least one real root
2) may have real roots
3) has at most one real root
4) has no real roots

43. If 2x + 3y = 7 and x – y = 1 are two normal of a parabola $y^2 = 4ax$ from a point then third normal may be

1) x – 3y +1 = 0 2) x + 3y = 5
3) x + 3y +1 = 0 4) x – 3y -7 = 0

44. If [.] denotes the greatest integer function then domain of the real valued function

$f(x)=\log_{[x+1/2]}\left|x^2+x-6\right|$ is

1) $\left(\frac{1}{2},1\right)\cup(1,\infty)$ 2) $\left[0,\frac{3}{2}\right]\cup(2,\infty)$ 3) $\left[\frac{3}{2},2\right)\cup(2,\infty)$ 4) $(0,1]\cup\left(\frac{3}{2},\infty\right)$

45. The length of sub tangent, ordinate and subnormal to the parabola $y^2 = 4ax$ at a point (different from origin) are in

1) AP 2) GP 3) HP 4) AGP

46. If $p,q\in R$ and $p^2+q^2-pq-p-q+1\le 0$ then $\begin{vmatrix} 1 & 2p & 3 \\ 3q & -1 & -5p \\ p & q & p \end{vmatrix}$ is equal to

A) 1 B) -1 C) -2 D) 0

47. The family of two curves represented by $\frac{dy}{dx}=\frac{x^2+x+1}{y^2+y+1}$ and $\frac{dy}{dx}+\frac{y^2+y+1}{x^2+x+1}=0$

1) Touch each other 2) are orthogonal

3) Are one and the same 4) Intersect at angle 45^0

48. The line $\frac{x-2}{3}=\frac{y+1}{2}=\frac{z-1}{-1}$ intersects the curve $xy=c^2, z=0$ if c is equal to

1) ± 1 2) $\pm\frac{1}{3}$ 3) $\pm\sqrt{5}$ 4) ± 2

49. Sum of first n terms of an AP (having positive terms) is given by $S_n=(1+2T_n)(1-T_n)$ (Where T_n is the nth term of the series). Then the value of $2T_1^2$ is

1) 2 2) 4 3) 1 4) 6

50. Let $f:R\to R^+$ be a differentiable function satisfying $f'(x)=2f(x)\ \forall x\in R$. Also $f(0)=1$ *and* $g(x)=f(x).\cos^2 x$. If n_1 represent number points of local maxima of $g(x)$ in $[-\pi,\pi]$ and n_2 is the number of points of local minima of $g(x)$ in $[-\pi,\pi]$ and n_3 is the number of points in $[-\pi,\pi]$ where $g(x)$ attains its global minimum value, then find the value of $(n_1+n_2+n_3)$

1)7 2) 6 3) 5 4)4

51. The locus of the middle points of chords of hyperbola $3x^2 - 2y^2 + 4x - 6y = 0$ parallel to $y = 2x$ is

1) $3x - 4y = 4$ 2) $3y - 4x + 4 = 0$

3) $4x - 4y = 3$ 4) $3x - 4y = 2$

52. If A and B represent the complex numbers z_1 and z_2 such that $|z_1 - z_2| = |z_1 + z_2|$ the circumcenter of ΔAOB , O being origin is

1) 0 2) $\frac{z_1 + 2z_2}{3}$ 3) $\frac{z_1 + z_2}{2}$ 4) $\frac{z_1 - z_2}{2}$

53. If $\angle A = 90^0$ in ΔABC, then $\tan^{-1}\left(\frac{c}{a+b}\right) + \tan^{-1}\left(\frac{b}{a+c}\right)$ is equal to

1) 0 2) 1 3) $\frac{\pi}{4}$ 4) $\frac{\pi}{6}$

54. The number of real values of "a" such that $a^2 - 2a \sin x + 1 = 0$ for any x, is

1) 1 2) 2 3) 3 4) 4

55. $\int \frac{(x^4 - x)^{1/4}}{x^5} dx$ is equal to

1) $\frac{4}{15}\left[1 - \frac{1}{x^3}\right]^{5/4} + c$ 2) $\frac{4}{5}\left[1 - \frac{1}{x^3}\right]^{5/4} + c$

3) $\frac{4}{15}\left[1 + \frac{1}{x^3}\right]^{5/4} + c$ 4) $\frac{4}{5}\left[1 + \frac{1}{x^3}\right]^{5/4} + c$

56. If x, y, z are angles of a triangle and tan x = 2 cot y and tan x + tan y + tan z = 6 then value of z is

1) $n\pi + \frac{\pi}{4}; n \in I$ 2) $n\pi + \tan^{-1} 2; n \in I$

3) $n\pi + \tan^{-1} 3; n \in I$ 4) $n\pi; n \in I$

57. The locus of the centre of the circle which touches the y-axis and also touches the circle $(x+1)^2 + y^2 = 1$ externally is

1) $\{(x, y) x^2 = 4y\} \cup \{(0, y) | y \le 0\}$

2) $\{(x, y) | y^2 = 4x\} \cup \{(x, 0) | x \le 0\}$

3) $\{(x, y) | x^2 + 4y = 0\} \cup \{(0, y) | y \ge 0\}$

4) $\{(x, y) | y^2 + 4x = 0\} \cup \{(x, 0) | x \ge 0\}$

58. Let $f:[a, b]\rightarrow[1, \infty)$ be a continuous function and let $g:\ \mathbb{R}\rightarrow\mathbb{R}$ be defined as

$$g(x)=\begin{cases} 0 & \text{if} \quad x<a, \\ \int_a^x f(t)dt & \text{if} \quad a\le x\le b, \\ \int_a^b f(t)dt & \text{if} \quad x>b. \end{cases}$$

Then

1) g(x) is continuous but not differentiable at 'a'

2) g(x) is differentiable on $\mathbb{R}$

3) g(x) is discontinuous at 'b'

4) g(x) is continuous and differentiable at either 'a' or 'b' but not both.

59. The area bounded by the curves $y = x^2 - 2x + 2$ and its inverse is given by

1) $\frac{1}{3}$ 2) $\frac{1}{6}$ 3) $\frac{5}{3}$ 4) $\frac{4}{3}$

60. Given a triangle with sides 17, 15 and 19 cm long. If a circle touching two smaller sides has its centre on the largest side, then the radius of the circle is

1) $\frac{51\sqrt{91}}{32}$ 2) $\frac{17\sqrt{91}}{32}$ 3) $\frac{51\sqrt{91}}{64}$ 4) $\frac{17\sqrt{91}}{64}$

CHEMISTRY

61. The herbicide 2,4-dichloro phenoxy acetic acid generally called 2,4-D is synthesized from 2, 4-dichloro phenol and chloroacetic acid. The synthesis involves------------ reaction.

1) S_N1 2) ArS_N 3) S_N2 4) S_Ni

62. Which of the following reactions is not feasible?

1) $(CH_3)_3\ CBr \xrightarrow{NaCN} (CH_3)_2\ C = CH_2 + NaBr + HCN$

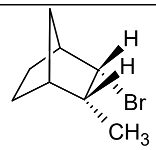
2) $R–CH_2–Cl \xrightarrow{NaHS} R - CH_2 - SH + NaCl$

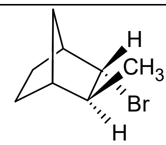
3) $R - CH_2 - SH_2^+ \xrightarrow{Br^-} R - CH_2 - Br + H_2S$

4) $R - CH_2 - OH_2^+ \xrightarrow{CN^-} R - CH_2–CN + H_2O$

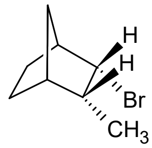
63. Identify the correct statement

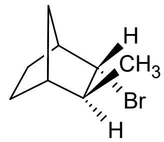
1) The rate of elimination of compound I is faster than II

I. 

II. 

2) The rate of elimination of compound II is faster than I

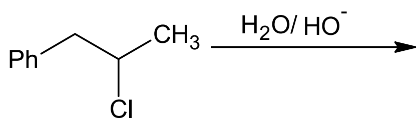
I. 

II. 

3) Elimination from non coplanar conformation is easier than anticoplanar conformation.

4) Elimination from syncoplanar conformation is easier than anticoplanar conformation.

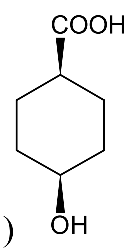
64. Identify the major product(s) in the following reaction?

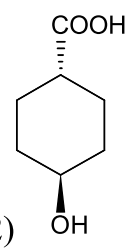
(R) 

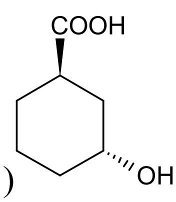
1) (S) $Ph\text{-}CH_2\text{-}CH(OH)\text{-}CH_3$ 2) (R) $Ph\text{-}CH_2\text{-}CH(OH)\text{-}CH_3$

3) $CH_3\text{-}CH(Ph)\text{-}CH_2OH$ 4) Both 2 & 3

65. Which of the following hydroxyl acids readily forms lactones?

1) 

2) 

3) 

4)All form lactones

66. Identify the correct statements regarding acid derivatives?

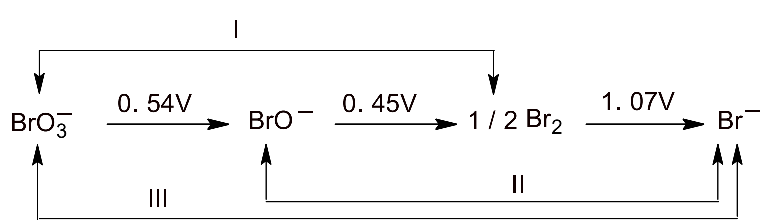
1) The correct order of resonance stabilisation is: Acid chloride > Amides > Esters

2) The correct order of resonance stabilisation is: Amides > Esters > Acid chlorides

3) The correct order of basic nature of oxygen is: Esters > Amides > Acid halides

4) The correct order of basic nature of oxygen is: Acid halides > Esters > Amides

67. The standard potentials at each stage for bromine species are shown in the following diagram.



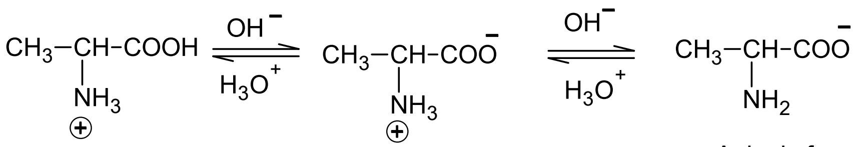
What is the value of ΔG^o for the reaction – I in the above diagram?

1) –251.86 kJ 2) –477.67 kJ 3) –208.44 kJ 4) –382.0 kJ

68. (S)-(–)-2-Methyl-1-butanol can be converted to (+)-2-Methylbutanoic acid without breaking any of the bonds to the asymmetric carbon. The configuration of (–)-2-Methylbutanoic acid is

1) S 2) R 3) S or R 4) Can't be determined

69. When a strongly acidic solution of alanine is titrating against a base the following equilibria are obtained



Cationic form Dipolar ion Anionic form

Given that pKa of cationic part is 2.3 and that of dipolar ion is 9.7. The incorrect statement from the following is?

1) Iso electric point of a Alanine is 6.0

2) When a pH of 2.3 is reached, half of the carboxylic acid proton are removed

3) When pH = 9.7 is reached one fourth of aminium groups protons are lost.

4) When a pH of 6.0 is reached all the alanine's carboxylic acid protons are lost.

70. Parke's process is used for De silverisation of lead, recovery of silver by parke's process is based on some facts. Pick out the in correct one from the following

1) Molten lead and zinc are nearly immiscible

2) Silver is less soluble in molten zinc than in molten lead

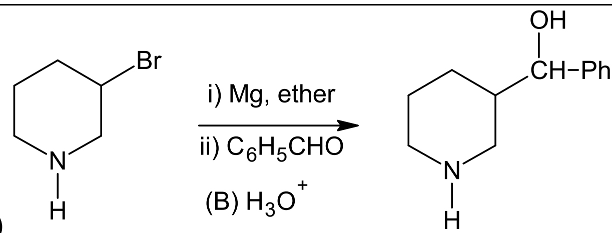
3) Zinc–Silver alloy solidifies earlier than molten lead

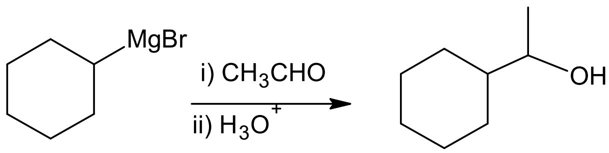
4) zinc being volatile can be separated from silver by distillation

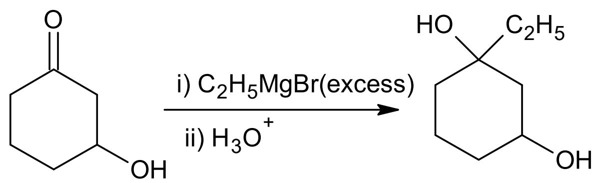
71. Oxydation state of 'V' is $Rb_4\,Na[HV_{10}O_{28}]$ is

1) + 5 2) +6 3) $+\frac{7}{5}$ 4) +4

72. Which of the following reaction is feasible ?

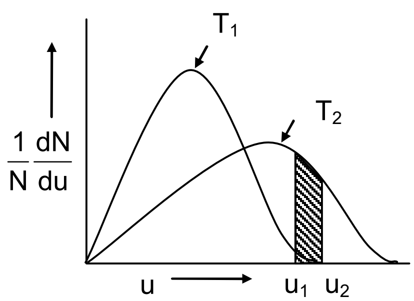
I) 

II) 

III) 

1) only II 2) I and II 3) II and III 4) all the three

73. Select the incorrect statement



1) At higher temp fraction of molecules with lower speed decreases

2) The speed distribution does not depend on mass of molecules

3) μ_{MP} at T_2 is higher than μ_{MP} at T_1

4) Shaded region in the plot shows the fraction of molecules in the speeds B/W U_1, and U_2

74. Inversion of sucrose $(C_{12}H_{22}O_{11})$ is first order reaction and is studied by measuring angle at rotation of different interval of time. $\underset{\text{sucrose}}{C_{12}H_{22}O_{11}} + H_2O \xrightarrow{H^+} \underset{\text{Glucose}}{C_6H_{12}O_6} + \underset{\text{Fructose}}{C_6H_{12}O_6}$

If $(r_\infty - r_0) = a$ and $(r_\infty - r_t) = a - x$ (Where r_0, r_t and r_∞ are the angle of rotation at the start, at the time t, and at the end of the reaction respectively, then there is 50%, inversion when

1) $r_0 = 2r_t - r_\infty$ 2) $r_0 = r_t - r_\infty$ 3) $r_0 - r_t - 2r_\infty$ 4) $r_0 = r_t + r_\infty$

75. 2×10^{-3} g of green algae absorbs 7×10^{-4} moles of CO_2 per hour by photosynthesis as per the following equation: $6nCO_2 + 5nH_2O \xrightarrow{h\gamma} (C_6H_{10}O_5)_n + 6nO_2$. If all the carbon of CO_2 is converted into starch, then how long will it take for the algae to increase its mass by 100%

1) 6.34 hours 2) 63.4 min 3) 6.34 min 4) 126.8 min

76. The work done in an open container at 300 K, when 112 g of iron reacts with dil. HCl to form $FeCl_2$ is

1) 1.2 k.cal 2) 4 k. Joul 3) 6 k.cal 4) 12 k.Joul

77. A big RED spherical balloon (radius = 6a) is filled up with gas. On this balloon six small GREEN_spherical balloons (radius = a) are stuck on the surface in a “specific” manner. As RED balloon is slowly deflated, a point comes when all these six **GREEN balloons** touch and **green balloons** arrange themselves in a 3-D closed packing arrangement. At that stage the radius of the **RED balloon** would have reduced by approximately.

1) 14.5 times 2) 1.414 times 3) 6.0 times 4) 2.42 times

78. In what order the reagents Na_2S, NaCl and NaI are added to an aqueous solution containing Ag^+, Cu^{+2} and Ni^{+2} ions in order to precipitate Ag^+ first Cu^{+2} second and Ni^{+2} last.

1) Na_2S, NaI, NaCl 2) NaCl, Na_2S, NaI

3) NaI, NaCl, Na_2S 4) NaCl, NaI, Na_2S

79. The volume percentage of Cl_2 at equilibrium in the dissociation of PCl_5 under a total pressure of 1.5 atm is (Kp=0.202),

1) 74.5 2) 36.5 3) 63.5 4) 25.5

80. $Ca_2B_6O_{11} + (A) + H_2O \longrightarrow Ca(HSO_3)_2 + (B)$

$(B) \xrightarrow[>200^0 C]{} (C) + H_2O$. Identify the correct statement about the reaction

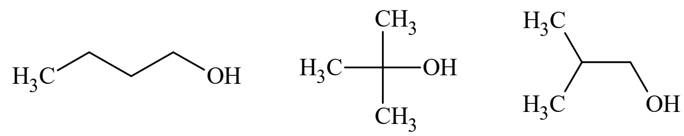
1) Both ‘A’ & ‘B’ are acidic A is more acidic than B

2) Aq.solution of ‘B’ liberates CO_2 from $NaHCO_3$

3) All transition metal oxides give coloured bead with ‘C’

4) All are correct

81. Arrange the following alcohols in order of decreasing K_b (molal elevation constant)



values. (A) (B) (C)

1) A > C > B 2) B > A > C 3) B > C > A 4) A =B=C

82. D – Mannose exists mainly in pyranose forms. The specific rotation of α-anomer is 29^0 and that of β-anomer is -17^0. The rotation the equilibrium mixture is $+14^0$. The % of α -anomer in the mixture is,

1) 32.6 2) 67.4 3) 42.6 4) 57.4

83. Match the tests with the respective salt solutions:

Column – I	Column – II
(A) White ppt. formation with aq. $BaCl_2$	(p) $BiCl_3$
(B) Add excess of HNO_3 and then Ammonium molybdate., yellow ppt	(q) $(CH_3COO)_2Mg$
(C) Add Conc.H_2SO_4, a pungent smell gas is evolved.	(r) $(NH_4)_3PO_4$
(D) Add Na_2HPO_4 with NH_4Cl and NH_4OH a white ppt.	(s) Na_2SO_4

1) A – r, B – s, C – p; D – q 2) A – r,s, B – r, C – p; D – q

3) A – p, B – q, C – r; D – s 4) A – r, B – r, s C – q; D – p

84. In which of the following pairs the later molecule is having more dipole moment than previous molecule

I. H_2O, NH_3 II. $CHCl_3, CH_3Cl$ III. NCl_3, NH_3 IV. HF, HCl

1) I, II 2) II, III 3) Only I 4) III, IV

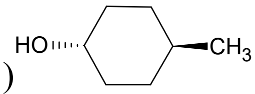
85.

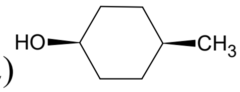
S.No.	Molecule(A)	Hybridization(B)	Shape(C)
1	IOF_3	SP^3d	Linear
2.	$XeOF_4$	SP^2	Tetrahedral
3.	Br_3^-	SP^3d^2	Sea saw
4.	NH_4^+	SP^3	Sq.pyramidal

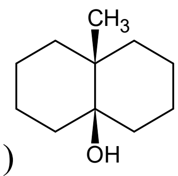
1) $A_1 - B_2 - C_3$; $A_3 - B_2 - C_1$ 2) $A_1 - B_4 - C_3$; $A_2 - B_4 - C_3$

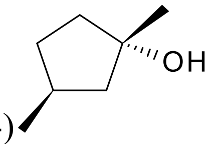
3) $A_3 - B_1 - C_1$; $A_4 - B_4 - C_2$ 4) $A_2 - B_1 - C_3$; $A_2 - B_3 - C_4$

86. Which of the following on dehydration does not give recemic mixture?

1) 

2) 

3) 

4) 

87. Which of the following statement is incorrect?

1) Silicates are wide spread in the earth crust because they are very insoluble in water

2) The $Si-O$, bonds are weaker than $C-O$ bonds because of bigger size of silicon atom

3) Zeolites are used as ion exchange materials, and as molecular sieve

4) Sharing of all four corners of a SiO_4 tetrahedron results in a three dimensional lattice of formula SiO_2.

88. Among the following, identify the correct statements.

1) The higher the gold number, the greater will be the protective power of a lyophilic colloid.

2) For an associated colloid, at critical micelle concentration osmotic pressure increases.

3) Surface tension of a lyophilic sol is less than that of its dispersion medium

4) In electrophoresis, only dispersion medium is allowed to migrate but not disperse phase.

89. Which of the following statements is incorrect?

1) The order of splitting energy is $PtCl_4^{2-} > PdCl_4^{2-} > NiCl_4^{2-}$

2) $[Co(NH_3)_6]^{3+}$ is colourless whereas $[Ni(H_2O)_6]^{2+}$ is coloured

3) $[M(en)(gly)]^{n+}$ will represent geometrical isomerism

4) The magnetic moment of $K_3[Fe(CN)_6]$ is $\sqrt{3}$ B.M

90. A 600 W mercury lamp emits monochromatic radiation of wavelength 331.3 nm. The number of photons emitted from the lamp per second is? $\left[h = 6.6260\times10^{-34}\ Js\right]$

1) 5×10^{20} 2) 1×10^{21} 3) 2×10^{22} 4) 2×10^{21}

KEY SHEET

1	**3**	2	**2**	3	**4**	4	**2**	5	**2**
6	**3**	7	**4**	8	**2**	9	**4**	10	**3**
11	**1**	12	**3**	13	**1**	14	**3**	15	**3**
16	**3**	17	**2**	18	**3**	19	**2**	20	**3**
21	**2**	22	**1**	23	**3**	24	**2**	25	**2**
26	**2**	27	**2**	28	**2**	29	**1**	30	**4**
31	**4**	32	**2**	33	**1**	34	**1**	35	**2**
36	**1**	37	**2**	38	**2**	39	**4**	40	**1**
41	**2**	42	**4**	43	**2**	44	**3**	45	**2**
46	**4**	47	**2**	48	**3**	49	**3**	50	**2**
51	**1**	52	**3**	53	**3**	54	**2**	55	**1**
56	**3**	57	**4**	58	**1**	59	**1**	60	**3**
61	**3**	62	**4**	63	**2**	64	**4**	65	**1**
66	**2**	67	**1**	68	**2**	69	**3**	70	**2**
71	**1**	72	**3**	73	**2**	74	**1**	75	**3**
76	**1**	77	**1**	78	**4**	79	**4**	80	**1**
81	**1**	82	**2**	83	**2**	84	**2**	85	**3**
86	**4**	87	**2**	88	**3**	89	**3**	90	**2**

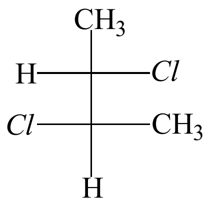
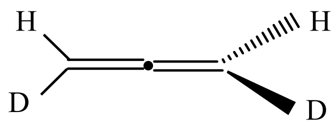
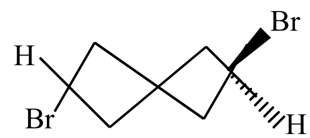
NARAYANA IIT ACADEMY INDIA

Sec: Sr. IIT **Date: 23-02-24**

Daily Practices Sheet - 04

CHEMISTRY

1. The chiral molecules are:

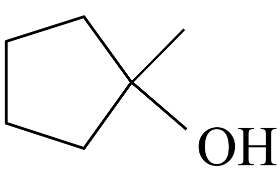
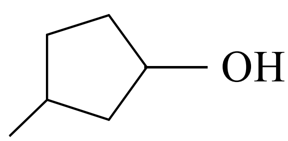
1)  2)  3) 

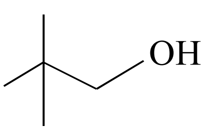
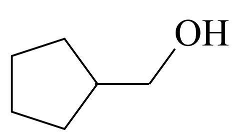
1)1, 2 2)2, 3 3)1, 3 4)1, 2, 3

2. A gas expands against a constant external pressure of 2.00 atm, increasing its volume by 3.40 L. Simultaneously, the system absorbs 400 J of heat from its surroundings. What is ΔE, in joules, for this gas?

1) – 689 2) – 289 3) + 400 4) + 289

3. Which alcohol can be prepared as single product by hydroboration – oxidation of an alkene.

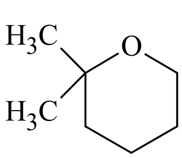
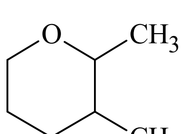
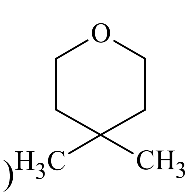
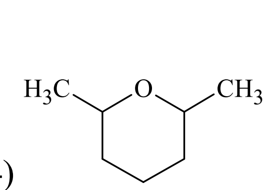
1)  2) 

3)  4) 

4. In the oxymercuration – demercuration of the following compound

$$H_2C{=}CH\text{-}\underset{}{\overset{CH_3}{\overset{|}{C}}}H\text{-}CH_2CH_2CH_2OH \xrightarrow[NaBH_4]{(CH_3COO)_2Hg} \text{Products}$$

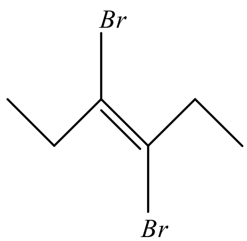
The major product is expected to be

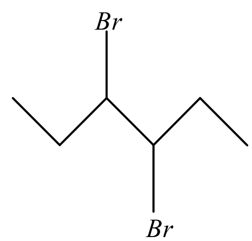
1)  2)  3)  4) 

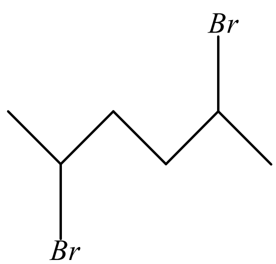
5. Four moles of PCl_5 are heated in a closed $4dm^3$ container to reach equilibrium at 400K. At equilibrium, 50% of PCl_5 is dissociated. What is the value of K_c for the dissociation of PCl_5 into PCl_3 *and* Cl_2 at 400K?

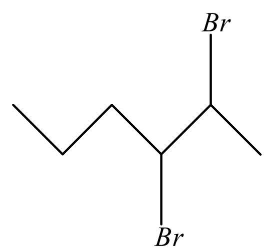
1) 0.50 2) 1.00 3) 1.25 4) 0.05

6. Compound 'X' on heating with Zn dust gives compound 'Y' which on treatment with O_3 followed by reaction with Zn dust gives propionaldehyde. The structure of 'X' is

1) 

2) 

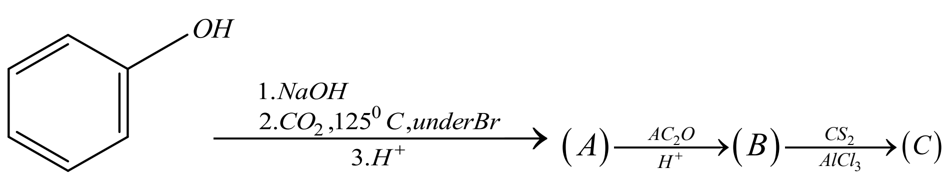
3) 

4) 

7. What is the volume of water needed to make normality of the solution 3N with respect to concentration of $\left(H^+\right)$, prepared by mixing 250 mL 6M HCl and 350 mL 8M HNO_3 ?

1) 833.3 mL 2) 933.3 mL 3) 1000 mL 4) 500 mL

8.



The correct statement/s about the above reaction is/are

I) (A) dissolves both in alkali and sodium carbonate

II) (B) gives color reaction with neutral $FeCl_3$

III)(C)exhibits tautomerism.

IV) (B) and (C) give effervescence with sodium carbonate

1) I, II, II 2) II, III, IV 3) I, III, IV 4) only IV

9. The hybridization of the central atoms in the following species is $\left[Ni(CN)_4\right]^{2-}$

1) dsp^2 2) sp^3 3) d^2sp^3 4) sp^2

10. Some of the following are essential amino acids

A: Valine, B: Leucine

C: Serine, D: Proline

E: Lysine, F: Aspartic acid

Select essential amino acid.

1) A, B, C 2) B, C, F 3) A, C, D 4) A, B, E

11. What is the similarity between buna – N and PHBV?

1) Both are copolymers

2) Both are biodegradable

3) Both have one same monomeric unit

4) Peroxide catalyst is used in their preparations

12. An organic compound $C_3H_9N(A)$ when treated with nitrous acid gave an alcohol and N_2 gas was evolved. (A) on warming with $CHCl_3$ and caustic potash gave (B) which on reduction gave isopropyl methyl amine. Predict the structure of (A).

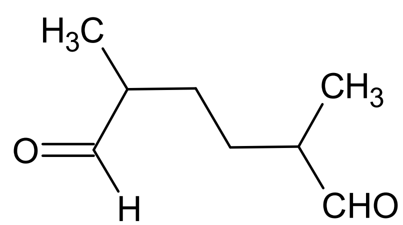
1) $CH_3-CH_2-CH_2-NH_2$

2) $\begin{matrix} CH_3 \\ CH_3 \end{matrix}\!\!\!>CH-NH_2$

3) $CH_3-CH_2-NH-CH_3$

4) $\begin{matrix} CH_3-N-CH_3 \\ | \\ CH_3 \end{matrix}$

13. A hydrocarbon C_8H_{14} adds one molecule of hydrogen. On ozonolysis the compound gives the given dialdehyde



The hydrocarbon is

1) 2, 5–dimethyl–3–hexene

2) 1–octene

3) 3, 6–dimethyl –1, 4–cyclohexadiene

4) 3, 6–dimethylcyclohexene

14. The correct expression for the vander wall's equation of states is –

1) $\left(p+a/n^2\ V^2\right)(V-nb)=nRT$ 2) $\left(p+an^2/V^2\right)(V-nb)=\Delta nRT$

3) $\left(p+an^2/V^2\right)(V-b)=nRT$ 4) $\left(p+an^2/V^2\right)(V-nb)=nRT$

15. For oxidation of iron

$4Fe(s)+3O_2(g)$ ® $2Fe_2O_3(s)$

$\Delta S_{system}=-549.4\ J/k-mole\ \text{at}\ 298\ k$

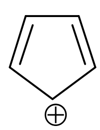
$\Delta H_{system}=-1648\ kJ/mole$

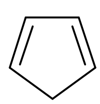
What is $\Delta S_{\text{universe, approximately}}$

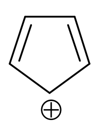
1) 4000 J/K-mole 2) 4981.6 J/K-mole

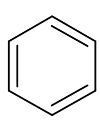
3) 5530.2 J/K-mole 4) 9000 J/K-mole

16. Identify the correct statement:

1)  is aromatic and paramagnetic

2)  is aromatic and diamagnetic.

3)  is antiaromatic and paramagnetic

4)  is antiaromatic and paramagnetic

17. The pH of 10^{-8} M HCl is

1) 7 2) 8 3) 7.02 4) 6.98

18. Tyndall effect is due to

1) reflection of light 2) scattering of light

3) absorption of light 4) adsorption of light

19. Number of significant figures in 1200 eggs is

1) 2 2) 3 3) 4 4) infinity

20. What transition in the hydrogen spectrum would have the same wavelength as the Balmer transition n =4 to n = 2 of He^{+} spectrum?

1) $n_1 = 1$ to $n_2 = 2$ 2) $n_1 = 2$ to $n_2 = 4$

3) $n_1 = 1$ to $n_2 = 3$ 4) $n_1 = 2$ to $n_2 = 3$

21. $CH_3COOH \xrightarrow{LiAlH_4} A$

$A + CH_3COOH \xrightarrow{H_3O^+} B + H_2O$

In the above reaction, 'A' and 'B' respectively are

1) $CH_3COOC_2H_5, C_2H_5OH$ 2) CH_3CHO, C_2H_5OH

3) C_2H_5OH, CH_3CHO 4) $C_2H_5OH, CH_3COOC_2H_5$

22. Which of the following has the minimum bond length?

1) O_2 2) O_2^+ 3) O_2^- 4) O_2^{2-}

23. 99% at a first order reaction was complete in 32 min. when will 99.9% of the reaction complete?

1) 48 min 2) 46 min 3) 50 min 4) 45 min

24. If a current of 0.5 amp flows through a metallic wire for 2 hrs, then how many electrons flow through the wire?

1) 2.24×10^{21} 2) 2.39×10^{21} 3) 2.246×10^{22} 4) 3.23×10^{22}

25. The edge length of a face centred cubic cell of an ionic substance is 508 pm. If the radius of the cation is 110 pm, the radius of the anion is

1) 288 pm 2) 398 pm 3) 618 pm 4) 144 pm

26. $N_2O_{4(g)}$ is 20% dissociated into $NO_{2(g)}$ at 27^0C and 760 torr. The density of equilibrium mixture is ______________

1) $3.11\frac{g}{L}$ 2) $7.33\frac{g}{L}$ 3) $4.8\frac{g}{L}$ 4) $8.2\frac{g}{L}$

27. In the extraction of copper, metal is formed in the Bessemer converter due to the reaction

1) $Cu_2S + 2Cu_2O \rightarrow 6Cu + SO_2$ 2) $Cu_2S \rightarrow 2Cu + S$

3) $Fe + Cu_2O \rightarrow 2Cu + FeO$ 4) $2Cu_2O \rightarrow 4Cu + O_2$

28. The correct order of first ionization energies of C, N, O and F is

1) C>N>O>F 2) C<N<O<F

3) C<O<N<F 4) C>O>N>F

29. Diamond is hard because:

1) all the four valence electrons are bonded to each carbon atoms by covalent bonds

2) it is a giant molecule

3) it is made up of carbon atoms

4) it cannot be burnt

30. Extra pure nitrogen can be obtained by heating:

1) NH_3 with CuO 2) NH_4NO_3 3) $(NH_4)_2Cr_2O_7$ 4) $Ba(N_3)_2$

PHYSICS

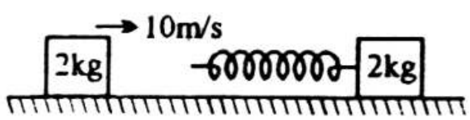
31. Specific resistance of a metal is determined experimentally using carey Foster's Bridge set up. One set of observations yielded the following readings Resistance $R = 54 \pm 2\Omega$, length of experimental wire $l = 136.0 \pm 1.0cm$, radius of wire $r = 0.26 \pm 0.02cm$. Then % error in specific resistance.

1)10.2% 2)16.4% 3)19.2% 4) 22.2%

32. The displacement of a body of mass 2 kg varies with time t as $s = t^2 + 2t$, where s is in metre and t is in second. The work done by all the forces acting on the body during the time interval $t = 2s$ to $t = 4s$ is

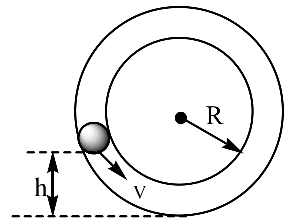
1) 36 J 2) 64 J 3) 100 J 4) 120 J

33. Block A of mass 2 kg moving with $10\ m/s$ strikes a spring of spring constant $\pi^2\ N/m$ attached to another 2 kg block B at rest kept on a smooth floor. The time for which rear moving block A remain in contact with spring will be



1) $\sqrt{2}$s 2) $\frac{1}{\sqrt{2}}$s 3) 1s 4) $\frac{1}{2}$s

34. With what minimum speed v must a small ball should be pushed inside a smooth vertical tube from a height h so that it may reach the top of the tube? Radius of the tube is R

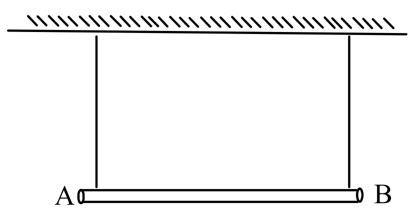


1) $\sqrt{2g(h+2R)}$ 2) $\frac{5}{2}R$ 3) $\sqrt{g(5R-2h)}$ 4) $\sqrt{2g(2R-h)}$

35. The centre of mass of a non-uniform rod of length L whose mass per unit length $\lambda = \frac{Kx^2}{L}$, where K is a constant and x is the distance from one end is

1) $\frac{3L}{4}$ 2) $\frac{L}{8}$ 3) $\frac{K}{L}$ 4) $\frac{3K}{L}$

36. A uniform rod of mass m and length l is suspended by means of two light inextensible strings as shown in figure. Tension in one string immediately after the other string is cut is

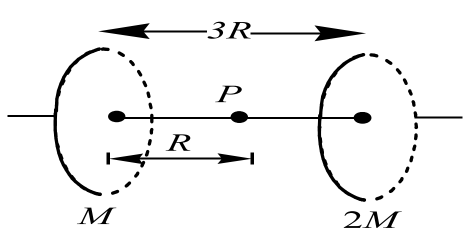


1) $\frac{mg}{2}$ 2) $2\ mg$ 3) $\frac{mg}{4}$ 4) mg

37. An inclined plane makes an angle of 60^0 with horizontal. A disc rolling down this inclined plane without slipping has a linear acceleration equal to

1) $\frac{g}{3}$ 2) $\frac{3}{4}g$ 3) $\frac{g}{\sqrt{3}}$ 4) $\frac{g}{2}$

38. Two rings having masses M and 2M, respectively, having the same radius R are placed coaxially as shown in the figure.



If the mass distribution on both the rings is non-uniform, then the gravitational potential at point P is

1) $-\frac{GM}{R}\left[\frac{1}{\sqrt{2}}+\frac{2}{\sqrt{5}}\right]$

2) $-\frac{GM}{R}\left[1+\frac{2}{2}\right]$

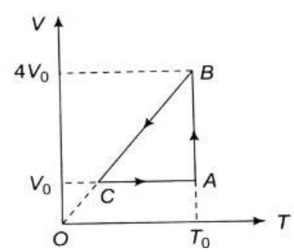
3) zero

4) cannot be determined from the given information

39. There is a horizontal film of soap solution. on it a thread is placed in the form of a loop. the film is pierced inside the loop and the thread becomes a circular loop of radius R. If the surface tension of the loop be T, then tension in the thread will be:

1) $\pi R^2 T$ 2) $2RT$ 3) RT 4) $\frac{\pi R^2}{T}$

40. One mole of an ideal gas in initial state A undergoes a cyclic process ABCA, as shown in the figure. Its pressure at A is P_0. Choose the correct option



1) Internal energy at B is greater than that at A

2) Work done by the gas in process $A \rightarrow B$ is P_0V_0

3) Pressure at C is $\frac{P_0}{4}$

4) Temperature at C is $\frac{T_0}{2}$

41. If C_P and C_v denote the specific heats of nitrogen gas per unit mass at constant pressure and constant volume respectively, then

1) $C_P - C_v = R/28$ 2) $C_P - C_v = R/14$

3) $C_P - C_v = R$ 4) $C_P - C_v = 28R$

42. A particle of mass 0.1 kg executes SHM under a force $F = (-10x)\,N$. Speed of particle at mean position is $6\,m/s$. Then amplitude of oscillations is

1) $0.6\,m$ 2) $0.2\,m$ 3) $0.4\,m$ 4) $0.1\,m$

43. A point source of sound is moving uniformly along a circle of radius 2 m in XY plane with centre at origin. A stationary detector is located at (0, 1). At time t = 0, the source is at (0, 2) and is going in clockwise direction with angular velocity $\frac{\pi}{6}$ rad / s. At what time the detector records maximum frequency?

1) $t = 2\sec$ 2) $t = 3\sec$ 3) $t = 9\sec$ 4) $t = 10\sec$

44. Two identical dipole $p\hat{i}$ and $p\hat{i}$ are placed at (a, 0, 0) and (-a, 0, 0) respectively. $\vec{E}$ is net electric field due to both the dipoles. Select **incorrect** statement

1) Electric flux crossing YZ-plane due to each dipole is zero

2) $\bar{E} = 0$ at $\left(0, \frac{\sqrt{3}a}{2}, \frac{\sqrt{5}a}{2}\right)$

3) Direction of $\bar{E}$ at $\left(0, \frac{3a}{2}, \frac{3a}{2}\right)$ is opposite to that at (0, 3a, 3a)

4) As one moves away from origin in YZ-plane, the magnitude of $\bar{E}$ decreases first and then increases and again decreases continuously

45. A parallel plate capacitor with air between the plates has a capacitance of 9pF. The separation between its plates is 'd'. The space between the plates is now filled with two dielectrics. One dielectric has dielectric constant $k_1 = 3$ and thickness $\frac{d}{3}$ while the other one has dielectric constant $k_2 = 6$ and thickness $\frac{2d}{3}$, capacitance of the capacitor is now

1) 40. 5pF 2) 20.25pF 3)1. 8 pF 4) 45 Pf

46. In a series L-R ac circuit, instantaneous potential difference across source and resistor are 10V and 5V respectively. At same instant, potential difference across inductor is denoted as V_L. Choose the **correct** statement

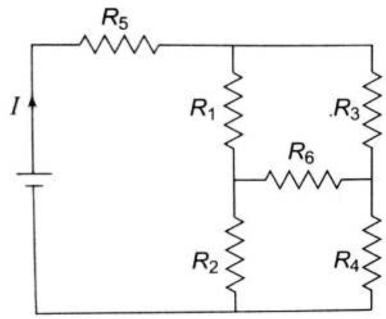
I) If $X_L = \sqrt{3}R$, V_L can be 5V II) If $X_L = \sqrt{3}R$, V_L can be 15V

III) If $X_L = R$, V_L can be 5V IV) If $X_L = R$, V_L can be 15V

1) I and III are only correct 2) I, II and III are only correct

3) I, III and IV are only correct 4) All are correct

47. In the given circuit it is observed that the current I is independent of the value of the resistance R_6. Then the resistance values must satisfy



1) $R_1R_2R_5 = R_3R_4R_6$

2) $\frac{1}{R_5}+\frac{1}{R_6}=\frac{1}{R_1+R_2}+\frac{1}{R_3+R_4}$

3) $R_1R_4 = R_2R_3$

4) $R_1R_3 = R_2R_4 = R_5R_6$

48. An electron and a proton are moving on straight parallel paths with same velocity. They enter a semi- infinite region of uniform magnetic field perpendicular to the velocity. Which of the following statements is true?

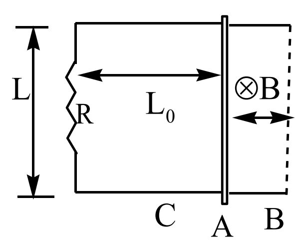
1) They will never come out of the magnetic field region

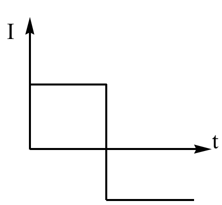
2) they will come out travelling along different inclined paths

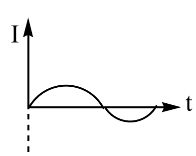
3) they will come out at the same time

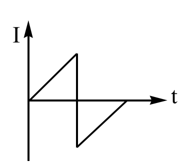
4) they will come out at different times

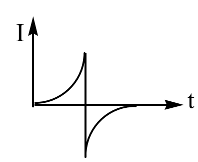
49. A conducting bar is moving along two parallel conducting bars, connected across a resistor R. The speed of the conducting bar is constant except at the turning points in uniform magnetic field B which is present every where. The conducting bar oscillates between positions B and C, starting from position A due to some external forces. The plot of induced current (I) through the rod against time (t) is best represented by



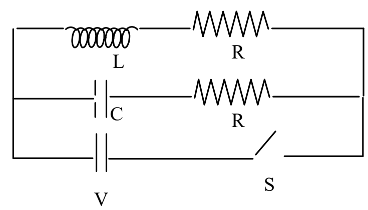
1) 

2) 

3) 

4) 

50. In the circuit shown in figure, $R=\sqrt{\frac{L}{C}}$. Switch S is closed at time $t=0$. The current through C and L would be equal after a time t equal to



1) CR 2) $CR\ln(2)$ 3) $\frac{L}{R\ln(2)}$ 4) LR

51. The amplitude of the electric field of a plane electromagnetic wave in air is $6.0\times10^{-4}Vm^{-1}$. The amplitude of the magnetic field will be

1) $1.8\times10^{5}T$ 2) $5.0\times10^{3}T$ 3) $2.0\times10^{-4}T$ 4) $2.0\times10^{-12}T$

52. A slit of width d is placed in front of a lens of focal length 0.5m and is illuminated with light of wavelength $5.89\times10^{-7}m$. The first diffraction minima on either side of the central diffraction maximum are separated by $2\times10^{-3}m$. The width d of the slit is (in m)

1) 2.045×10^{-4} 2) 2.105×10^{-4} 3) 2.945×10^{-4} 4) 0.125×10^{-4}

53. One slit of a double slit experiment is covered by a thin glass plate of refractive index 1.4 and the other by a thin glass plate of refractive index 1.7. The point on the screen where the central maximum fell before the glass plates were inserted is now occupied by what had been the fifth bright fringe before. Assume the plates to have the same thickness and light of wavelength 480 nm, find the value of thickness of each plate.

1) 2.4 μm 2) 4.8 μm 3) 8 μm 4) 16 μm

54. Two identical glass $(\mu_g=3/2)$ equiconvex lenses of focal length f are kept in contact. The space between the two lenses is filled with water $(\mu_w=4/3)$. The focal length of the combination is:

1) f 2) $\frac{f}{2}$ 3) $\frac{4f}{3}$ 4) $\frac{3f}{4}$

55. The potential energy of a particle of mass m is given by

$$V(x)=\begin{cases}E_0 & ,0\le x\le 1\\ 0 & ,x>1\end{cases}$$

λ_1 and λ_2 are the de-Broglie wavelengths of the particle, when $0 \le x \le 1$ and $x > 1$ respectively. If the total energy of particle is $2E_0$, then λ_1 / λ_2 is

1) 2 2) $\sqrt{2}$ 3) 1 4) 4

56. ${}_{92}U^{235}$ is α active. Then in a large quantity of the element

I) The probability of a nucleus disintegrating during one second remains constant for all time

II) The energy of the emitted α particle is less than the disintegration energy of U^{235} nucleus

1) I and II are true 2) I and II are false

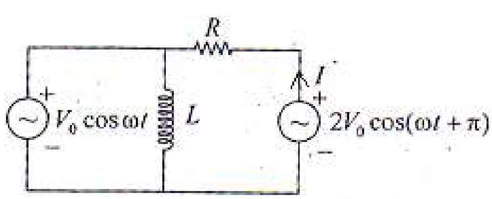
3) I is true and II is false 4) I is false and II is true

57. Energy liberated in the de-excitation of hydrogen atoms from 3rd level to 1st level falls on a photo – cathode. Later when the same cathode is exposed to a spectrum of some unknown hydrogen like gas, excited to 2nd energy level it is found that the de – Broglie wave length of the fastest photo electrons now ejected has decreased by a factor of ‘3’. For this new gas, difference of the (energy gap of 3rd energy level and 1st energy level) and (energy gap of 3rd energy level and 2nd energy level) is found to be 3 times the ionization energy of hydrogen atom. Then

1) The gas is Helium 2) Work function of photo cathode is 8.5eV

3) Both 1 and 2 are true 4) Both 1 and 2 are wrong

58. The diagram shows an AC circuit with two voltage sources of same frequency. Find out the value of current I shown in the fig.



1) $I = \left(\frac{V_0}{R}\right)\sin\omega t$ 2) $I = \left(\frac{V_0}{R}\right)\cos\left(\omega t + \pi/2\right)$

3) $I = \left(\frac{V_0}{R}\right)\cos\omega t$ 4) $I = \left(\frac{-3V_0}{R}\right)\cos\omega t$

59. Two full turns of the circular scale of screw gauge cover a distance of 1 mm on its main scale. The total number of divisions on the circular scale is 50. Further, it is found that the screw gauge has a zero error of -0.03 mm. While measuring the diameter of a thin wire, a student notes the main scale reading of 3 mm and the number of circular scale division in line with the main scale as 35. The diameter of the wire is

1) 3.38 mm 2) 3.32 mm 3) 3.73 mm 4) 3.67 mm

60. In a meter bridge experiment, when two resistors 6Ω and 20Ω are connected in left gap and right gap respectively, zero deflection position is obtained at 20 cm from left end of bridge wire. When the resistors are interchanged, zero deflection position shifted towards right by 56 cm. End corrections at left and right ends are denoted by e_L and e_R respectively. Then

1) $e_L = 0,\ e_R = 1cm$ 2) $e_L = 4cm,\ e_R = 1cm$

3) $e_L = 2cm,\ e_R = 1cm$ 4) $e_L = 4cm,\ e_R = 0$

MATHS

61. If $2f(x)+xf\left(\frac{1}{x}\right)-2f\left(\left|\sqrt{2}\sin\pi\left(x+\frac{1}{4}\right)\right|\right)=4\cos^2\left(\frac{\pi x}{2}\right)+x\cos\frac{\pi}{x},\forall x\in R-\{0\}$ then which of the following statement is incorrect?

1) $f(2)+f\left(\frac{1}{2}\right)=1$ 2) $f(2)+f(1)=0$

3) $f(2)+f(1)=f\left(\frac{1}{2}\right)$ 4) $f(1)f\left(\frac{1}{2}\right)f(2)=1$

62. Let k be a non zero real number. If $f(x)=\begin{cases}\dfrac{(e^x-1)^2}{\sin\left(\frac{x}{k}\right).\log\left(1+\frac{x}{4}\right)} & ;x\neq 0\\ 12 & ;x=0\end{cases}$

Is a continuous function, then the value of k

1) 1 2) 2 3) 3 4) 4

63. If f is differentiable function satisfying $f\left(\frac{1}{n}\right)=0$, for all $n\ge 1, n\in I$, then

1) $f(x)=f'(x)=0, x\in(0,1]$

2) $f'(0)=0=f(0)$

3) $f(0)=0$, but $f'(0)$ not necessarily zero

4) $|f(x)|\le 1, x\in(0,1]$

64. $\int \frac{\cos^3 x+\cos^5 x}{\sin^2 x+\sin^4 x}dx$ equals to

1) $\sin x-6\tan^{-1}(\sin x)+c$

2) $\sin x-2\sin^{-1}x+c$

3) $\sin x-2(\sin x)^{-1}-6\tan^{-1}(\sin x)+c$

4) $\sin x-2(\sin x)^{-1}-5\tan^{-1}(\sin x)+c$

65. Let $f:R\to R$ and $g:R\to R$ be respectively given by $f(x)=|x|+1$ and $g(x)=x^2+1$. Define $h:R\to R$ by $h(x)=\begin{cases}\max\{f(x),g(x)\} & if\ x\le 0\\ \min\{f(x),g(x)\} & if\ x>0\end{cases}$ Then number of points at which $h(x)$ is not differentiable is

1) 0 2) 4 3) 2 4) 3

66. $\underset{n\to\infty}{Lt}\frac{(1^2+2^2+......+n^2)(1^3+2^3+........+n^3)}{1^6+2^6+.....+n^6}=$

1) $\frac{5}{12}$ 2) $\frac{7}{12}$ 3) $\frac{9}{5}$ 4) $\frac{12}{7}$

67. Let I be the purchase value of an equipment and $V(t)$ be the value after it has been used for t years. The value $V(t)$ depreciates at a rate given by differential equation $\frac{dV(t)}{dt}=-K(T-t)$ where $K>0$ is a constant and T is the total life in years of the equipment. Then the scrap value $V(T)$ of the equipment is:

1) $I-\frac{KT^2}{2}$ 2) $I-\frac{K(T-t)^2}{2}$ 3) e^{-KT} 4) $T^2-\frac{I}{K}$

68. The distance from the origin, to the normal to the curve $x = 2\cos t + 2t\sin t, y = 2\sin t - 2t\cos t$ at $t = \frac{\pi}{4}$ is

1) 4 2) $2\sqrt{2}$ 3) 2 4) $\sqrt{2}$

69. Equation of plane containing two parallel lines $\frac{x+1}{3} = \frac{y-2}{2} = \frac{z}{1}$ and $\frac{x-3}{3} = \frac{y+4}{2} = \frac{z-1}{1}$ is

1) $8x + y - 26z + 6 = 0$ 2) $8x + y - 26z + 9 = 0$

3) $3x + 2y + z - 4 = 0$ 4) $x + y + z = 1$

70. A ray of light through $A(1,2,3)$ and strikes mirror plane $x + y + z = 12$ at B and reflected ray pass through $C(3,5,9)$ then B=

1) $(-7,0,19)$ 2) $(7,0,-19)$ 3) $(1,2,3)$ 4) $(7,0,19)$

71. The number of values of k, for which the system of equations:

$(k+1)x + 8y = 4k$

$kx + (k+3)y = 3k - 1$

Has no solution, is

1) infinite 2) 1 3) 2 4) 3

72. Let t be a complex number such that $|t| = 1$, then $z = \frac{at+b}{t-c}$ describes,

(where a and b are also complex numbers)

1) a circle as t varies if $|c| \neq \frac{1}{2}$ 2) a line as t varies if $|c| = 1$

3) a circle as t varies if $|c| = 1$ 4) None of these

73. If 'a' is real constant and A, B, C are variable angles and $\left(\sqrt{a^2-4}\right)\text{TanA} + a\text{TanB} + \left(\sqrt{a^2+4}\right)$ $\text{TanC} = 6a$ then the least value of $\text{Tan}^2\text{A} + \text{Tan}^2\text{B} + \text{Tan}^2\text{C}$ is____

1) 10 2) 12 3) 14 4) 20

74. Let $\vec{f}(t) = [t]\vec{i} + \{t - [t]\}\vec{j} + [t+1]\vec{k}$, where [.] denotes the greatest integer function. Then the vectors $\vec{f}\left(\frac{5}{4}\right)$ *and* $\vec{f}(t), 0 < t < 1$ are

1) perpendicular to each other
2) parallel to each other
3) inclined at an angle $\cos^{-1}\frac{2}{\sqrt{5(1+t^2)}}$
4) inclined at $\cos^{-1}\frac{8+t}{9\sqrt{(1+t^2)}}$

75. Let $\vec{p}, \vec{q}, \vec{r}$ be three mutually perpendicular vectors with same magnitude. If $\vec{x}$ satisfies the relation $\vec{p}\times\{(\vec{x}-\vec{q})\times\vec{p}\}+\vec{q}\times\{(\vec{x}-\vec{r})\times\vec{q}\}+\vec{r}\times\{(\vec{x}-\vec{p})\times\vec{r}\}=\vec{0}$, then $\vec{x}=$

1) $\frac{1}{3}(\vec{p}+\vec{q}+\vec{r})$ 2) $\frac{1}{2}(\vec{p}+\vec{q}+\vec{r})$ 3) $\frac{1}{2}(\vec{p}+\vec{q}-2\vec{r})$ 4) $\frac{1}{2}(2\vec{p}+\vec{q}-\vec{r})$

76. A square of side a lies above the x-axis and has one vertex at the origin. The side passing through the origin makes an angle $\alpha\left(0<\alpha<\frac{\pi}{4}\right)$ with the positive direction of x-axis. The equation of its diagonal not passing through the origin is

1) $y(\cos\alpha+\sin\alpha)+x(\cos\alpha-\sin\alpha)=a$
2) $y(\cos\alpha-\sin\alpha)-x(\sin\alpha-\cos\alpha)=a$
3) $y(\cos\alpha+\sin\alpha)+x(\sin\alpha-\cos\alpha)=a$
4) $y(\cos\alpha+\sin\alpha)+x(\sin\alpha+\cos\alpha)=a$

77. Let $f:[-2,3]\to[0,\infty)$ be a continuous function such that $f(1-x)=f(x)$ for all $x\in[-2,3]$.

If R_1 is the numerical value of the area of the region bounded by $y=f(x), x=-2, x=3$ and the axis of x and $R_2=\int_{-2}^{3}xf(x)\,dx$, then

1) $3R_1=2R_2$ 2) $2R_1=3R_2$ 3) $R_1=R_2$ 4) $R_1=2R_2$

78. $px+qy=40$ is a chord of minimum length of circle $(x-10)^2+(y-20)^2=729$. If the chord passes through $(5, 15)$, then $p^{2013}+q^{2013}$ is equal to

1) 0 2) 2 3) 2^{2013} 4) 2^{2014}

79. The sum of all the solutions of the equation $8\cos x\left(\cos\left(\frac{\pi}{6}+x\right).\cos\left(\frac{\pi}{6}-x\right)-\frac{1}{2}\right)=1\ in\left[0,\pi\right]$ is.

1) $\frac{13\pi}{9}$ 2) $\frac{11\pi}{9}$ 3) $\frac{16\pi}{9}$ 4) $\frac{17\pi}{9}$

80. Let $\lambda x-y+7=0$ is any chord of the parabola $x^2=28y$ meeting at A and B and the tangents drawn to the parabola at A, B meet at C. Then locus of circumcentre of ΔABC is

1) $x^2=14(y-7)$ 2) $x^2=14y-7$ 3) $x^2=14y+7$ 4) $x^2=14-y$

81. A differentiable function f(x) is such that $f(x)=\sin x-\int_1^{x^2}\frac{f\left(\sqrt{t}\right)}{t}dt; x\geq 1$

Then, If $a,b\in N$ & the value of $f(\pi)=\frac{a(\sin b-\cos b-\pi)}{\pi^2}$; where $a+b=$ _____

1) 2 2) 0 3) 3 4) 1

82. A curve posses a property that the slope of the tangent of its point is proportional to the square of the abscissa of the point of tangency. If the equation of the tangent to the curve at the point $(1,1)$ is $x-y=0$, then the equation of the curve is

1) $y=x^2$ 2) $y=2x^3-1$ 3) $3y=2x^3+1$ 4) $3y=x^3+2$

83. A variable tangent to the circle $x^2+y^2=1$ intersects the ellipse $\frac{x^2}{4}+\frac{y^2}{2}=1$ at points P and Q. The locus of the point of intersection of tangents to the ellipse at P and Q is another ellipse. If e_1 and e_2 are the eccentricities of the two ellipses then e_1e_2 equal to

1) $\frac{1}{\sqrt{2}}$ 2) $\frac{\sqrt{6}}{4}$ 3) $\frac{\sqrt{3}}{4}$ 4) 1

84. The number of ways in which six men and five women can dine at a round table if no two women are to sit together is given by

1) $6!\times 5!$ 2) 30 C) $5!\times 4!$ 4) $7!\times 5!$

85. How many different words can be formed by jumbling the letters in the word MISSISSIPPI in which no two S are adjacent?

1) $8\times {}^6C_4\times {}^7C_4$ 2) $6\times 7\times {}^8C_4$ 3) $6\times 8\times {}^7C_4$ 4) $7\times {}^6C_4\times {}^8C_4$

86. The number of integers greater than 6,000 that can be formed, using the digits 3, 5, 6, 7 and 8, without repetition

1) 216 2) 192 3) 120 4) 72

87. Let α and β are the roots of the Quadratic equation $x^2-1154x+1=0$ then the value of $\sqrt[4]{\alpha}+\sqrt[4]{\beta}$ is

1)1 2) 4 3) 6 4) 10

88. If the sum of n terms of the series $\frac{2n+1}{2n-1}+3\left(\frac{2n+1}{2n-1}\right)^2+5\left(\frac{2n+1}{2n-1}\right)^3+.........$ is 36 then the value of n is

1) 5 2) 4 3) 6 4) 7

89. If $f(x)=x^4-2x^3+3x^2-ax+b$ is a polynomial such that when it is divided by $(x-1)$ and $(x+1)$ the remainder are 5 and 19 respectively. If $f(x)$ is divided by $(x-2)$, the remainder is

1) 0 2) 5 3) 10 4) 2

90. If the coefficients of r^{th}, $(r+1)^{th}$, and $(r+2)^{th}$ terms in the binomial expansion of $(1+y)^m$ are in A.P., then m and r satisfy the equation

1) $m^2-m(4r-1)+4r^2-2=0$ 2) $m^2-m(4r+1)+4r^2+2=0$

3) $m^2-m(4r+1)+4r^2-2=0$ 4) $m^2-m(4r-1)+4r^2+2=0$

KEY SHEET

1	**2**	2	**2**	3	**4**	4	**2**	5	**1**
6	**2**	7	**1**	8	**3**	9	**3**	10	**4**
11	**1**	12	**2**	13	**4**	14	**4**	15	**2**
16	**3**	17	**4**	18	**2**	19	**4**	20	**1**
21	**4**	22	**2**	23	**1**	24	**3**	25	**4**
26	**1**	27	**1**	28	**3**	29	**1**	30	**4**
31	**3**	32	**2**	33	**3**	34	**4**	35	**1**
36	**3**	37	**3**	38	**1**	39	**2**	40	**3**
41	**1**	42	**1**	43	**4**	44	**3**	45	**1**
46	**2**	47	**3**	48	**4**	49	**1**	50	**3**
51	**4**	52	**3**	53	**3**	54	**4**	55	**2**
56	**1**	57	**3**	58	**4**	59	**1**	60	**4**
61	**4**	62	**3**	63	**2**	64	**3**	65	**4**
66	**2**	67	**1**	68	**3**	69	**1**	70	**1**
71	**2**	72	**2**	73	**2**	74	**4**	75	**2**
76	**1**	77	**4**	78	**4**	79	**1**	80	**1**
81	**3**	82	**4**	83	**2**	84	**1**	85	**4**
86	**2**	87	**3**	88	**2**	89	**3**	90	**3**